

A.N. Kolmogorov and approximation theory

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A.N. Kolmogorov and approximation theory

V.M. Tikhomirov

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Andrei Nikolaevich Kolmogorov published very few papers concerned with approximation theory. This area of mathematics was on the periphery of his scientific interests. Yet his influence on the formation and development of approximation theory has turned out to be immense. It gave rise to a number of new scientific trends which represent a whole new stage in the development of approximation theory (one of three from the point of view now generally accepted). More than a hundred researchers have been involved in the development of this stage, more than a thousand articles are known to have been published on the subject, more than thirty doctoral dissertations have been defended, and several scientific schools have been founded.

The aim of this article is to take a look at the development of a scientific area that has grown up from seeds apparently scattered by the wayside.

I feel it is my duty to express my thanks to G.G. Magaril-II'yaev for help in the preparation of this article.

§1. Kolmogorov's work on approximation theory

In this section we shall set out the most important results from six articles by Kolmogorov ([1], Nos. 27, 28, 40, 49, [2], No. 31, [3], No. 7).

Today we usually distinguish three periods in approximation theory. The first is the Chebyshev period, beginning in 1854, when methods of

approximation to *specific functions by polynomials and rational functions* began to be investigated. The second period is usually reckoned from the work of Bernstein and Jackson (1911–1912), concerned with the relation between the *smoothness of functions and their best approximations by polynomials*. Beginning in the 1930's (largely under Kolmogorov's influence) this theme was somewhat changed and the approximation of *classes of functions by polynomials* (algebraic and trigonometric) began to be developed. The third period was directly related to Kolmogorov's own work. Here the *best methods of approximation to classes of functions began to be considered*; that is, the classical methods (algebraic polynomials, rational functions, trigonometric polynomials) were compared with *any other means of approximation* from a given family of approximating sets described by the same number of parameters. This led Kolmogorov to introduce new concepts—*diameter* and *entropy*. Here Kolmogorov himself repeatedly turned his attention to the use of an *exact* solution of the given problems, which also had major consequences.

In 1962, speaking at the International Congress in Stockholm, Kolmogorov suggested a programme of still more intensive study of individual functions and collections of functions, proposing to take account of the *complexity*, as well as the accuracy, of an algorithm that reproduces a function with given accuracy (see [3] No. 8). It may well be that this work will mark the beginning of a new fourth stage of approximation theory.

Beside all this, Kolmogorov laid the foundations of several special areas. To mention three: inequalities for derivatives on the line and half-line, criteria for the elements of best approximation of functions, ε -entropy in unit time of classes of functions on the line. This list is far from exhaustive. Several of the ideas we have spoken about have found applications (in particular in Kolmogorov's own work) to problems not directly related to approximation theory, but grouped around one of the central problems of mathematical analysis: *how complex are functions of several variables?*

After these general remarks, let us turn to the analysis of particular papers by Kolmogorov.

1.1. Estimates for the remainders of Fourier series.

This article, [1] No. 27, is dated 30 November 1934, and was published in 1935. It is a reply to Lebesgue's article of 1910, where he proves that if a function $x(\cdot)$ satisfies a Lipschitz condition, then the asymptotic relation $\|x(\cdot) - S_n x(\cdot)\|_{C(T)} \asymp n^{-1} \log n$ holds, where $S_n x(\cdot)$ is the n th partial sum of the Fourier series of $x(\cdot)$, and $\alpha_n \asymp \beta_n$ means that there exist $C_1 > 0$, $C_2 > 0$, and $n_0 \in \mathbb{N}$ such that $C_1 \alpha_n \leq \beta_n \leq C_2 \alpha_n$ for all $n \geq n_0$.

Kolmogorov transformed Lebesgue's formulation of the problem in the following way. First he introduced the following class of functions on the

circle $\mathbf{T}$ ($r \in \mathbf{N}$):

$$W_{\infty}^r(\mathbf{T}) = \{x(\cdot) \mid x^{(k)}(\cdot) \in C(\mathbf{T}), \quad 0 \leq k \leq r-1, \quad x^{(r-1)}(\cdot) \in AC(\mathbf{T}), \\ \|x^{(r)}(\cdot)\|_{L_{\infty}(\mathbf{T})} \leq 1\}.$$

(Here $\mathbf{T}$ is realized as the interval $[-\pi, \pi]$, where π and $-\pi$ are identified, $C(\mathbf{T})$ is the space of continuous functions, and $AC(\mathbf{T})$ the space of absolutely continuous functions on $\mathbf{T}$.⁽¹⁾)

Secondly, he posed the problem of determining the precise location in the given class of the greatest possible deviation of the function $x(\cdot) \in W_{\infty}^r(\mathbf{T})$ from its n th Fourier partial sum. That is, he posed the exact extremal problem of finding the quantity

$$\varphi_{nr} = \sup \{ \|x(\cdot) - S_n x(\cdot)\|_{L_{\infty}(\mathbf{T})} \mid x(\cdot) \in W_{\infty}^r(\mathbf{T}) \}.$$

In [1] No. 27 he proved the following:

Theorem 1. *Let r be a natural number.*

a) if r is odd, then $\varphi_{nr} = \pi^{-1} \int_{\mathbf{T}} \left| \sum_{k \geq n+1} k^{-r} \sin kt \right| dt$;

b) $\varphi_{nr} = 4\pi^{-2} n^{-r} \log n + O(n^{-r})$.

$\triangleleft 0$. In the proof Kolmogorov used the following well-known representation of $x(\cdot) - S_n x(\cdot)$ as a convolution:

$$(1) \quad x(\cdot) \in W_{\infty}^r(\mathbf{T}) \Rightarrow x(t) - S_n x(t) = (-1)^{r+1} \pi^{-1} \int_{\mathbf{T}} D_{nr}(t-\tau) x^{(r)}(\tau) d\tau,$$

where

$$D_{n2m}(t) = (-1)^{m+1} \sum_{k \geq n+1} k^{-2m} \cos kt, \\ D_{n2m+1}(t) = (-1)^{m+1} \sum_{k \geq n+1} k^{-(2m+1)} \sin kt.$$

The following well-known relations are also used.

$$\text{Let } D_n(t) = 2^{-1} + \sum_{k=1}^n \cos kt \text{ (the Dirichlet kernel), } F_n(t) = (n+1)^{-1} \sum_{k=0}^n D_k(t)$$

(the Fourier kernel). Then

$$(2) \quad F_n(t) \geq 0, \quad \pi^{-1} \int_{\mathbf{T}} F_n(t) dt = 1, \quad \pi^{-1} \int_{\mathbf{T}} |D_n(t)| dt = \\ = 4\pi^{-2} \log(n+1) + O(1).$$

⁽¹⁾Similar classes can be defined for any p :

$$W_p^r(\mathbf{T}) = \{x(\cdot) \mid x^{(k)}(\cdot) \in C(\mathbf{T}), \quad 0 \leq k \leq r-1, \quad x^{(r-1)}(\cdot) \in AC(\mathbf{T}), \quad \|x^{(r)}(\cdot)\|_{L_p(\mathbf{T})} \leq 1\}.$$

They are usually called *Sobolev classes*.

1. Let us prove a). Let $r = 2m + 1$. The function $D_{n2m+1}(\cdot)$ is odd (see (1)), so that the integral over a period of $\text{sign } D_{n2m+1}(\cdot)$ is zero, and hence there is a function $\bar{x}(\cdot)$, zero in mean, and such that $\bar{x}^{(r)}(\cdot) = \text{sign } D_{n2m+1}(\cdot)$. But then

$$\bar{x}(t) \stackrel{(1)}{=} \pi^{-1} \int_{\mathbb{T}} D_{n2m+1}(t-\tau) \text{sign } D_{n2m+1}(\tau) d\tau \stackrel{\text{def}}{\in} W_{\infty}^{2m+1}(\mathbb{T}),$$

from which we obtain

$$\begin{aligned} \pi^{-1} \int_{\mathbb{T}} \left| \sum_{k \geq m+1} k^{-r} \sin kt \right| dt &\stackrel{(1)}{=} \pi^{-1} \int_{\mathbb{T}} |D_{n2m+1}(\tau)| d\tau = \\ &= \left| \pi^{-1} \int_{\mathbb{T}} D_{n2m+1}(\tau) \text{sign } D_{n2m+1}(\tau) d\tau \right| \stackrel{(1)}{=} |\bar{x}(0) - S_n \bar{x}(0)| \stackrel{\text{def}}{\leq} \\ &\leq \sup \left\{ \max_t \left| \pi^{-1} \int_{\mathbb{T}} D_{n2m+1}(t-\tau) x^{(r)}(\tau) d\tau \right| \mid |x^{(r)}(t)| \leq 1 \right\} \leq \\ &\leq \pi^{-1} \int_{\mathbb{T}} |D_{n2m+1}(t)| dt \stackrel{(1)}{=} \pi^{-1} \int_{\mathbb{T}} \left| \sum_{k \geq n+1} k^{-r} \sin kt \right| dt. \end{aligned}$$

2. Let us prove b) for even $r = 2m$. By applying the Abel transformation (Ab) twice to D_{n2m} , using the notations $\Delta k = k^{-2m} - (k-1)^{-2m}$, $\Delta^2(k) = k^{-2m} - 2(k-1)^{-2m} + (k-2)^{-2m}$, we obtain

$$\begin{aligned} (3) \quad D_{n2m}(t) &\stackrel{(1)}{=} (-1)^{m+1} \sum_{k \geq n+1} k^{-2m} \cos kt \stackrel{(\text{Ab})}{=} (-1)^{m+1} (-(n+1)^{-2m} D_n(t) + \\ &+ \sum_{k \geq n+1} \Delta(k) D_k(t)) \stackrel{(\text{Ab}), (2)}{=} (-1)^{m+1} (-(n+1)^{-2m} D_n(t) - \\ &- (n+1) \Delta(n+1) F_n(t) + \sum_{k \geq n+1} (k+1) \Delta^2(k) F_k(t)) = \\ &= (-1)^m (n+1)^{-2m} D_n(t) + R(t). \end{aligned}$$

Hence, using (2) and the obvious relations $\Delta(k) \asymp k^{-(2m+1)}$, $\Delta^2(k) \asymp k^{-(2m+2)}$, we have

$$\begin{aligned} (4) \quad \int_{\mathbb{T}} |R(t)| dt &\leq (n+1) \Delta(n+1) + \sum_{k \geq n+1} (k+1) \Delta^2(k) \ll (n+1)^{-2m}, \\ \varphi_{n2m} &\stackrel{(1)}{\leq} \pi^{-1} \int_{\mathbb{T}} |D_{n2m}(t)| dt \stackrel{(3)}{\leq} \\ &\stackrel{(3)}{\leq} \pi^{-1} \left(\int_{\mathbb{T}} (n+1)^{-2m} |D_n(t)| dt + \int_{\mathbb{T}} |R(t)| dt \right) \stackrel{(2), (4)}{\leq} \\ &\stackrel{(2), (4)}{\leq} 4\pi^{-2} n^{-2m} \log n + O(n^{-2m}). \end{aligned}$$

It is easy to show that the lower bound is attained by the function $\tilde{x}(\cdot)$:

$$\tilde{x}^{(r)}(t) = \{\text{sign} \sin(2n+1)t/2, t \notin [0, 2\pi/(2n+1)]\}$$

and

$$0, t \in [0, 2\pi/(2n+1)]. \quad \triangleright$$

1.2. Diameters.

This article, [1] No. 28, is dated 1 May 1935, and was published in 1936. Here there are no references to predecessors. Kolmogorov introduces the quantity called the *Kolmogorov diameter* or the *Kolmogorov n -diameter*.⁽¹⁾

Let X be a normed space with unit ball BX , and let $C \subset X$ be a centrally-symmetric subset; the *Kolmogorov n -diameter* of C in X is the quantity

$$d_n(C, X) := \inf d(C, L_n, X) := \inf \sup_{x \in C} \inf_{y \in L_n} \|x - y\|,$$

where the lower bound is taken over all n -dimensional subspaces of X .

A subspace $\hat{L}_n$ is called *extremal* if $d_n(C, X) = d(C, \hat{L}_n, X)$.

In approximation theory, approximations to functions (of given smoothness) have been studied, by say the space $\mathcal{T}_n$ of trigonometric polynomials of degree $\leq n$. This suggests the question: for the same collection of functions cannot we find another subspace of the same dimension $(2n+1)$ which approximates the given set (class) better? This problem also leads to the concept of diameter.

We should remark that a similar quantity, called the diameter, was first introduced in dimension theory by Uryson in 1922, and other diameters (also related to the concept of dimension of a set) were introduced by Aleksandrov in 1933. The idea of the Aleksandrov diameter is very close to that of Kolmogorov's, but Kolmogorov does not pursue this analogy in his article.

Kolmogorov introduces the class ($r \in \mathbf{N}$)

$$W_2^r(\mathbf{T}) = \{x(\cdot) \mid x^{(k)}(\cdot) \in C(\mathbf{T}), 0 \leq k \leq r-1, x^{(r-1)}(\cdot) \in AC(\mathbf{T}), \\ \|x^{(r)}(\cdot)\|_{L_2(\mathbf{T})} \leq 1\},$$

and proves the following result.

Theorem 2. *Let r and n be natural numbers. Then*

$$d_{2n-1}(W_2^r(\mathbf{T}), L_2(\mathbf{T})) = d_{2n}(W_2^r(\mathbf{T}), L_2(\mathbf{T})) = n^{-r}, \quad d_0 = \infty,$$

where the space $\mathcal{T}_{n-1}$ of trigonometrical polynomials of degree $\leq n-1$ is an extremal $(2n-1)$ -dimensional space.

⁽¹⁾Translator's note. Also called *n-width*.

◁ 0. The proof uses the following obvious and well-known facts:

$$a) \quad x(\cdot) \in W_2^r(\mathbf{T}) \Rightarrow x(t) = a_0 \cdot 2^{-1} + \sum_{k \in \mathbf{N}} (a_k \cos kt + b_k \sin kt),$$

$$(1) \quad \|x(\cdot)\|_{L_2(\mathbf{T})}^2 = a_0^2 \cdot 2^{-1} + \sum_{k \in \mathbf{N}} (a_k^2 + b_k^2),$$

$$\|x^{(r)}(\cdot)\|_{L_2(\mathbf{T})}^2 = \sum_{k \in \mathbf{N}} k^{2r} (a_k^2 + b_k^2);$$

b) if X is a Hilbert space, L_s and L_{s+1} are s - and $(s+1)$ -dimensional subspaces respectively, then $L_s^\perp \cap L_{s+1} \neq \{0\}$ ($L^\perp$ is the orthogonal complement of L).

1. Upper bound.

Let $\hat{L}_{2n-1} = \mathcal{T}_{n-1} = \text{lin} \{\sin k\cdot, \cos k\cdot, 0 \leq k \leq n-1\}$,

$$y(t) = S_{n-1}x(t) = a_0 \cdot 2^{-1} + \sum_{k=1}^{n-1} (a_k \cos kt + b_k \sin kt).$$

Then (where Id is the identity transformation)

$$\begin{aligned} (2) \quad \|x(\cdot) - y(\cdot)\|_{L_2(\mathbf{T})}^2 &\stackrel{(1)}{=} \sum_{k \geq n} (a_k^2 + b_k^2) \stackrel{\text{Id}}{=} \\ &= n^{-2r} \sum_{k \geq n} n^{2r} (a_k^2 + b_k^2) \stackrel{k \geq n}{\leq} n^{-2r} \sum_{k \geq n} k^{2r} (a_k^2 + b_k^2) \leq \\ &\leq n^{-2r} \sum_{k \in \mathbf{N}} k^{2r} (a_k^2 + b_k^2) \stackrel{(1)}{=} n^{-2r} \|x^{(r)}(\cdot)\|_{L_2(\mathbf{T})}^2 \stackrel{\text{def}}{\leq} n^{-2r}. \end{aligned}$$

2. Lower bound.

Let L_{2n} be the $2n$ -dimensional subspace $L_2(\mathbf{T})$. Let $\bar{L}_{2n+1} = \mathcal{T}_{n+1} = \text{lin} \{\sin k\cdot; \cos k\cdot; 0 \leq k \leq n\}$. Let $x(\cdot) \in \bar{L}_{2n+1} \cap n^{-r} BL_2(\mathbf{T})$. Then by (1) we obtain

$$\begin{aligned} (3) \quad x(t) &= a_0 \cdot 2^{-1} + \sum_{k=1}^n (a_k \cos kt + b_k \sin kt), \\ \|x(\cdot)\|_{L_2(\mathbf{T})}^2 &\stackrel{(1)}{=} a_0^2 \cdot 2^{-1} + \sum_{k=1}^n (a_k^2 + b_k^2) \stackrel{\text{def}}{\leq} n^{-2r} \Rightarrow \\ &\Rightarrow \|x^{(r)}(\cdot)\|_{L_2(\mathbf{T})}^2 \stackrel{0.1}{=} \sum_{k=1}^n k^{2r} (a_k^2 + b_k^2) \stackrel{(k \leq n)}{\leq} \sum_{k=1}^n n^{2r} (a_k^2 + b_k^2) \stackrel{(3)}{\leq} 1. \end{aligned}$$

Hence $\bar{L}_{2n+1} \cap n^{-r} BL_2(\mathbf{T}) \subset W_2^r(\mathbf{T})$, and then from (0.b) there is a function $\bar{x}(\cdot) \in L_{2n}^\perp \cap \bar{L}_{2n+1}$ such that $\|\bar{x}(\cdot)\|_{L_2(\mathbf{T})} = n^{-r}$, and so, since the diameters are monotonic ($C_1 \subset C_2 \Rightarrow d_k(C_1, X) \leq d_k(C_2, X)$):

$$\begin{aligned} (4) \quad d(W_2^r(\mathbf{T}), L_{2n}, L_2(\mathbf{T})) &\geq d(\bar{L}_{2n+1} \cap n^{-r} BL_2(\mathbf{T}), L_{2n}, L_2(\mathbf{T})) \geq \\ &\geq (\text{since } \bar{x}(\cdot) \in L_{2n}^\perp) \|\bar{x}(\cdot)\|_{L_2(\mathbf{T})} = n^{-r}. \end{aligned}$$

Because of the inequality $d_{2n} \leq d_{2n-1}$ we finally obtain

$$n^{-r} \leq \inf_{L_{2n}} d(W_2^r(\mathbf{T}), L_{2n}, L_2(\mathbf{T})) \stackrel{\text{def}}{=} d_{2n}(W_2^r(\mathbf{T}), L_2(\mathbf{T})) \leq \\ \leq d_{2n-1}(W_2^r(\mathbf{T}), L_2(\mathbf{T})) \stackrel{\text{def}}{=} d(W_2^r(\mathbf{T}), \hat{L}_{2n-1}, L_2(\mathbf{T})) \stackrel{(2)}{\leq} n^{-r}. \triangleright$$

In the same article Kolmogorov also considered the non-periodic case. Let $W_2^r(I)$ denote the class of functions on the interval $I = [0, 1]$, with continuous k th derivatives for $0 \leq k \leq r-1$, and absolutely continuous $(r-1)$ th derivative, $\|x^{(r)}(\cdot)\|_{L_2(I)} \leq 1$. Kolmogorov computed the n -diameter of $W_2^r(I)$ in $L_2(I)$, using the following result of Krein. Consider the equation of Sturm-Liouville type

$$Lx = x^{2r}, \quad Lx + (-1)^r \lambda^2 x = 0, \quad x^{(k)}(0) = x^{(k)}(1) = 0, \quad r \leq k \leq 2r-1.$$

Krein proved that there is a sequence $\{\lambda_k\}_{k \geq r}$, $0 < \lambda_r < \lambda_{r+1} < \dots$ of eigenvalues of the operator L and a corresponding sequence of eigenfunctions $\{x_k(\cdot)\}_{k \geq r}$ which: a) when augmented by a system $\{x_k(\cdot)\}_{k=0}^{r-1}$ of orthonormal polynomials of degree k , $0 \leq k \leq r-1$, becomes a complete orthonormal system in $L_2(I)$; b) the system $\{x_k^{(r)}(\cdot)\}_{k \geq r}$ is a complete orthogonal system in $L_2(I)$ and $\|x_k^{(r)}(\cdot)\|_{L_2(I)} = \lambda_k$.

By repeating the argument of Theorem 1 word for word with $\hat{L}_{2n-1}$ replaced by $\hat{L}_n = \text{lin}\{x_k(\cdot)\}_{k=0}^{n-1}$ and $\bar{L}_{2n+1}$ by $\bar{L}_{n+1} = \text{lin}\{x_k(\cdot)\}_{k=0}^n$, we arrive at the following result.

Theorem 2'. *Let r be a natural number. Then*

$$d_n(W_2^r(I), L_2(I)) = \lambda_n^{-1}, \quad n \geq r, \quad d_n = \infty, \quad 0 \leq n \leq r-1.$$

Thus in the first case the classical space $\mathcal{T}_n (= \hat{L}_{2n+1})$ of trigonometrical polynomials turns out to be extremal. (Kolmogorov mistakenly supposed it to be unique.) In the second case it turns out that the extremal spaces are different for various r , and the classical space of algebraic polynomials can be shown not to be extremal. In his article Kolmogorov drew the attention of researchers to the problem of finding an *optimal approximating apparatus*.

1.3. On inequalities for intermediate derivatives on the line.

This article, [1] No. 40, was published in 1939, preceded by an announcement in Comptes Rendus in 1938.

The author refers to the following result of Hadamard (1914): if $x(\cdot)$ is a bounded differentiable function and $\dot{x}(\cdot)$ satisfies a Lipschitz condition on the line, then

$$(1) \quad \|\dot{x}(\cdot)\|_{C^b(\mathbf{R})} \leq 2 \|x(\cdot)\|_{C^b(\mathbf{R})}^{1/2} \|\ddot{x}(\cdot)\|_{L_\infty(\mathbf{R})}^{1/2}$$

($C^b(\mathbf{R})$ is the space of bounded continuous functions with norm

$$\|x(\cdot)\|_{C^b(\mathbf{R})} = \sup_{t \in \mathbf{R}} |x(t)|).$$

In the late 1930's Kolmogorov set his student Shilov the problem of extending Hadamard's inequality to the case of n -times differentiable functions. Shilov, having proved a number of partial results, formulated the general hypothesis that the extremal functions in the problem were the functions that had recently occurred in the work of Favard [4], and Akhiezer and Krein [5] (and still earlier in Euler; Kolmogorov refers to the article [5] by Akhiezer and Krein, which we shall have cause to return to again).

Shilov was unable to substantiate his hypothesis fully. Kolmogorov himself was carried away by the problem and solved it. According to S.M. Nikol'skii's memoirs, Kolmogorov was very proud of this result.

Theorem 3 (Kolmogorov's inequality for intermediate derivatives on the line). *If $n \geq 2$ is a natural number, $x(\cdot) \in W_{\infty}^n(\mathbf{R})$, and $^{(1)} 0 < k < n$, then the following inequality holds:*

$$(1') \quad \|x^{(k)}(\cdot)\|_{C^b(\mathbf{R})} \leq K_{n-k} K_n^{\frac{n-k}{n}} \|x(\cdot)\|_{C^b(\mathbf{R})}^{\frac{n-k}{n}} \|x^{(n)}(\cdot)\|_{L_{\infty}(\mathbf{R})}^{\frac{k}{n}},$$

where $K_r = 4\pi^{-1} \sum_{h \in \mathbf{N}} (-1)^{(r+1)(h+1)} (2h-1)^{-(r+1)}$ (the Favard constants), and

equality is attained for the 2π -periodic function $x_n(\cdot)$ for which

$x_n^{(n)}(t) = \text{sign} \sin t$, $\int_{\mathbf{T}} x_n(t) dt = 0$. (These are called the Euler functions or splines.)

This result led to a whole area in the theory of extremal problems and classical analysis (inequalities for intermediate derivatives on $\mathbf{R}$ and $\mathbf{R}_+$), which turned out to be essential in approximation theory.

We shall give a proof of the Kolmogorov inequalities for $k < n-1$ due to the American mathematician Cavaretta.

$\triangleleft 0$. The following properties of the functions $x_n(\cdot)$ (see the statement of the theorem) play a central part in the proof:

a) $\|x_n^{(k)}(\cdot)\|_{C(\mathbf{T})} = K_{n-k}$; b) $x_n(\cdot)$ on any interval $[-s\pi, s\pi]$ has $2s$ -alternance, that is, it takes the values K_n $2s$ times with successively alternating signs.

1. Lemma. *Let s be a natural number, let $x(\cdot) \in W_{\infty}^n(\mathbf{R})$ and be periodic with period $2\pi s$ (together with its derivatives up to order $r-1$), and $\|x(\cdot)\|_{C([-s\pi, s\pi])} \leq K_n$. Then $\|x^{(k)}(\cdot)\|_{C([-s\pi, s\pi])} \leq K_{n-k}$.*

Proof of the lemma. Suppose there is a function $\bar{x}(\cdot)$, satisfying the condition of the lemma, but such that the point zero is the maximum of $\bar{x}^{(k)}(\cdot)$ and $\bar{x}^{(k)}(0) > K_{n-k}$. By a translation we can arrange that the function $\bar{x}_n(t) := x_n(t + \beta)$ has the property $\bar{x}_n^{(k)}(0) = K_{n-k}$. We put

(1) $x(\cdot) \in W_{\infty}^n(\mathbf{R}) \Leftrightarrow x^{(k)}(\cdot) \in C^b(\mathbf{R})$, $0 \leq k \leq n-1$, $x^{(n-1)}(\cdot) \in AC(\mathbf{R})$, $x^{(n)}(\cdot) \in L_{\infty}(\mathbf{R})$.

$\alpha := \bar{x}^{(k)}(0)/K_{n-k}$, $z(t) = \bar{x}(t) - \alpha x_n(t)$. Then it is clear that $\alpha > 1$,

$$z^{(k)}(0) = z^{(k+1)}(0) = 0.$$

By 0 b) $P(z(\cdot))$, the number of sign changes of $z(\cdot)$ on $[-s\pi, s\pi]$, is not less than $2s$, so by Rolle's theorem, $P(z^{(k)}(\cdot)) \geq 2s$, and also $z^{(k)}(0) = z^{(k+1)}(0) = 0$ whence (by Rolle's theorem) the number of zeros of $z^{(k+1)}(\cdot)$ is greater than $2s$, and by Rolle's theorem again the number of zeros of $z^{(n-1)}(\cdot)$ is greater than $2s$. But $z^{(n-1)}(\cdot)$ is strictly monotonic on each of the intervals $[k\pi + \beta, (k+1)\pi + \beta]$, $|k| \leq s$ (since $|\alpha \bar{x}_n^{(n)}(t)| \equiv \alpha > 1$ on each of these intervals from the definition of $x_n(\cdot)$, and $|x_n^{(n)}(t)| \leq 1$), that is, $P(z^{(n-1)}(\cdot)) \leq 2s$. This contradiction proves the lemma.

2. Suppose there is a function $\hat{x}(\cdot) \in W_\infty^n(\mathbf{R})$ such that $\|\hat{x}(\cdot)\|_{C^b(\mathbf{R})} \leq K_n$, $\|\hat{x}^{(k)}(\cdot)\|_{C^b(\mathbf{R})} = \gamma > K_{n-k}$. By a translation and multiplication by -1 we can arrange that $\hat{x}^{(k)}(0) > K_{n-k}$.

Let $y(\cdot)$ denote an infinitely differentiable function on $\mathbf{R}$ with support in $[-1, 1]$, vanishing identically on $[-1/2, 1/2]$, $0 \leq y(t) \leq 1$. We put $x_s(t) = \hat{x}(t)y(t/s)$. Then by Leibniz's formula we have

$$(2) \quad x_s^{(n)}(t) = \sum_{j=0}^n C_n^j \hat{x}^{(j)}(t) y^{(n-j)}(t/s) s^{-(n-j)}.$$

It follows from (2) that there is a number δ and numbers δ_k and δ_n such that

$$(3) \quad \|\hat{x}_s^{(n)}(\cdot)\|_{L_\infty(\mathbf{R})} \leq 1 + \delta_n, \quad \hat{x}_s^{(k)}(0) > \gamma(1 - \delta_k), \\ (\gamma(1 - \delta_k)/(1 + \delta_n)) > K_{n-k}.$$

Here clearly $\|\hat{x}_s(\cdot)\|_{C^b(\mathbf{R})} \leq K_n$. Let $\xi(\cdot) = \hat{x}_s(\cdot)(1 + \delta_n)^{-1}$. Then the function $\xi(\cdot) \in W_\infty^n(\mathbf{R})$ and it can be periodically extended with period $2\pi s$ to the whole of $\mathbf{R}$ (since $\text{supp } \xi = [-\pi s, \pi s]$), $\|\xi(\cdot)\|_{C^b(\mathbf{R})} \leq K_n$, and $\xi^{(k)}(0) > K_{n-k}$. But this contradicts the lemma. $\triangleright$

1.4. Diameters of the octahedron.

This article, [2] No. 31, written by Kolmogorov together with Petrov and Smirnov, appeared in 1947. In it the authors ran into a geometric problem in approximation. Seven years later in 1954 Stechkin noticed that by using the methods of this article and a related article by Mal'tsev (cited in [2] No. 31) an exact formula for the diameter of an n -dimensional octahedron in l_2^n could be derived. Stechkin in turn applied this result on the diameter of the octahedron to the problem of the diameters of classes of functions, and the outcome was that the octahedron theorem and Stechkin's result gave rise to new areas of research concerned with finding the diameters of finite-dimensional sets and classes of functions of various degrees of smoothness.

Let l_p^n , $1 \leq p \leq \infty$, denote the Banach space $1 \leq p < \infty$,

$$\{x \in \mathbf{R}^n \mid x = (x_1, \dots, x_n), \|x\|_{l_p^n} = \left(\sum_{h=1}^n |x_h|^p \right)^{1/p}, \|x\|_{l_\infty^n} = \max_{1 \leq i \leq n} |x_i|\},$$

and let Bl_p^n be the unit ball in l_p^n . Thus $Bl_1^n = \left\{x \mid \sum_{h=1}^n |x_h| \leq 1\right\}$ is an n -dimensional octahedron. We then have:

Theorem 4 (Kolmogorov-Petrov-Smirnov-Mal'tsev-Stechkin theorem on the diameters of octahedra in l_2). *Let k and n be natural numbers, $0 \leq k < n$. Then*

$$d_k(Bl_1^n, l_2^n) = \left(\frac{n-k}{n} \right)^{1/2}.$$

◁ 0. The proof uses the following known facts.

a) If $\{f_i\}_{i=1}^h$ is an orthonormal system in $\mathbf{R}^n$, then

$$(1) \quad d^2(x, \text{lin}\{f_i\}_{i=1}^h, l_2^n) = \|x\|_{l_2^n}^2 - \sum_{s=1}^h (\langle x, f_s \rangle)^2,$$

where we recall that for a normed space X and a subspace L

$$d(x, L, X) = \inf\{\|x - y\| \mid y \in L\},$$

and for $x, y \in \mathbf{R}^n$

$$\langle x, y \rangle = \sum_{h=1}^n x_h y_h.$$

b) If A is an orthogonal $n \times n$ matrix, then l_2^n decomposes into the direct sum of one- and two-dimensional orthogonal subspaces invariant with respect to A .

1. Lower bound.

Let L_k be a k -dimensional subspace of $\mathbf{R}^n$, and $\{f_i\}_{i=1}^h$ be an orthonormal basis of L_k . Then

$$(2) \quad d^2(x, L_k, l_2^n) \stackrel{(0a)}{=} \|x\|_{l_2^n}^2 - \sum_{s=1}^h (\langle x, f_s \rangle)^2,$$

from which, letting $\{e_i\}_{i=1}^n$ denote the standard basis in $\mathbf{R}^n$ ($e_1 = (1, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0)$ and so on) and putting $f_{is} = \langle f_i, e_s \rangle$, we obtain

$$(3) \quad \sum_{s=1}^n d^2(e_s, L_k, l_2^n) \stackrel{(2)}{=} n - \sum_{i=1}^h \sum_{s=1}^n f_{is}^2 = n - k.$$

since $\sum_{s=1}^n f_{is}^2 \stackrel{\text{def}}{=} \|f_i\|_{l_2^n}^2 \stackrel{\text{def}}{=} 1$. From (3) and the fact (i) that the greatest distance from a point of the octahedron to a subspace is attained at one of

its vertices, we have

$$(3') \quad d^2(Bl_1^n, L_k, l_2^n) \stackrel{\text{def}}{=} \max_{x \in Bl_1^n} d^2(x, L_k, l_2^n) \stackrel{(i)}{=} \max_{1 \leq s \leq n} d^2(e_s, L_k, l_2^n) \geq \\ \geq n^{-1} \sum_{s=1}^n d^2(e_s, L_k, l_2^n) \stackrel{(3)}{=} \frac{n-k}{n}. \quad (1)$$

Since L_k is arbitrary, (3') implies that $d_k^2(Bl_1^n, l_2^n) \geq (n-k)n^{-1}$. It is also clear from (3') that if $\hat{L}_k$ is equidistant from all the vertices e_s , then

$$(4) \quad d(Bl_1^n, \hat{L}_k, l_2^n) = \left(\frac{n-k}{n} \right)^{1/2}.$$

2. Upper bound.

From (4) it is sufficient to prove that there is a k -dimensional subspace of l_2^n equidistant from all the e_s . This, in actual fact, was done by Mal'tsev, who gave an explicit construction of $\hat{L}_k$. Following Ismagilov we shall merely prove an existence theorem.

Let T denote the orthogonal transformation defined by $Te_i = e_{i+1}$, $i = 1, \dots, n-1$, $Te_n = e_1$.

Clearly the one-dimensional subspace $L_1 = \text{lin}\{e_1 + \dots + e_n\}$ is invariant under T (that is, $TL_1 = L_1$). Now consider all the two-dimensional orthogonal subspaces for T (see Ob). If from them we can build up a space of dimension k or $k-1$, then the problem is solved (in the second case we adjoin L_1). If however we can only build up a space of dimension less than $k-1$, then the remaining spaces are one-dimensional, and with their help we can again "build up" a k -dimensional subspace. Thus let $\hat{L}_k$ be a k -dimensional invariant subspace. Letting Px denote the orthogonal projection of x on $\hat{L}_k$, and putting $Pe_i = \xi_i$, $T^{-1}\xi_i = \xi'_i$, we have:

$$d(e_i, \hat{L}_k, l_2^n) \stackrel{\text{def}}{=} \|e_i - Pe_i\| \stackrel{\text{def}}{=} \|Te_{i-1} - \\ - T(T^{-1}\xi_i)\| \stackrel{(Ob)}{=} \|e_{i-1} - \xi'_i\| \stackrel{(\xi'_i \in \hat{L}_k)}{\geq} \inf_{\xi \in \hat{L}_k} \|e_{i-1} - \xi\| \stackrel{\text{def}}{=} d(e_i, \hat{L}_k, l_2^n).$$

The opposite inequality is obtained similarly, and hence from (4) the upper bound. $\triangleright$

1.5. Kolmogorov's criterion for the closest element in the complex case.

In this article, [1] No. 49, published in 1948, he proved a result of this kind.

Theorem 5. Let L_n be an n -dimensional subspace of the space $C(T, \mathbf{C})$ of continuous complex functions on a compact space T with the uniform

⁽¹⁾The method used here is called the *averaging method*.

metric, and let $x(\cdot) \in C(T, \mathbf{C}) \setminus L_n$. Then an element $\hat{y}(\cdot) \in L_n$ is closest to $x(\cdot)$ if and only if for any element $y(\cdot) \in L_n$ we have

$$\min_{t \in T_0} \operatorname{Re}(\overline{\hat{y}(t)}(\hat{y}(t) - x(t))) \leq 0, \text{ where}$$

$$T_0 = \{t \in T \mid \|x(t) - \hat{y}(t)\| = \|x(\cdot) - \hat{y}(\cdot)\|_{C(T, \mathbf{C})}\}.$$

We shall not prove this theorem here, since later we shall prove a far-reaching generalization which is no more difficult to obtain. We shall discuss this theorem then.

1.6. ε -entropy.

This article, [3] No. 7, was written by Kolmogorov with the present author and published in 1959.⁽¹⁾ It contains proofs of a series of results of Kolmogorov, published earlier in *Uspekhi Matematicheskikh Nauk* (1955), the proceedings of the All-Union Mathematical Congress (1956), and *Doklady Akademii Nauk SSSR* (1956). The last article is entitled "On some asymptotic characteristics of completely bounded metric spaces". To the same series belong an article [1] No. 57, on the linear dimension of topological vector spaces, and articles concerned with the solution of Hilbert's 13th problem. We shall touch on these themes later in the appropriate section, but here we shall only discuss the definitions, the statement of the problem, and formulate some theorems.

In the 1950's new reasons arose for Kolmogorov to return to approximation theory again. On the one hand there was his study of Vitushkin's work on Hilbert's 13th problem, and on the other his enthusiasm for Shannon's information theory. All this induced Kolmogorov to introduce a new concept— ε -entropy—into approximation theory.

Let (X, d) be a metric space, $C \subset X$ a subset, and $N_\varepsilon(C, X)$ the smallest number N for which there are N points $\{x_i\}_{i=1}^N$ forming an ε -net for C , that is, for any $x \in C$ there is an $i = i(x)$, $1 \leq i \leq N$, such that $d(x, x_i) \leq \varepsilon$. The logarithm to base 2 of the number $N_\varepsilon(C, X)$ was denoted by $\mathcal{H}_\varepsilon(C, X)$ and called the ε -entropy. It is easy to see the analogy between ε -entropy and n -diameter. The function $N \rightarrow \varepsilon_N(C, X)$, inverse to $\varepsilon \rightarrow N_\varepsilon(C, X)$, can be defined similarly to the diameter:

$$\varepsilon_N(C, X) = \inf_{x \in C} \sup_i \inf d(x, x_i),$$

where the lower bound is taken over all subsets $\{x_1, \dots, x_N\}$ of N elements.

The diameters d_n are related to the study of the approximation properties of n -dimensional subspaces, while the diameters ε_n are concerned with the approximation properties of sets of n points.

⁽¹⁾= Amer. Math. Soc. Transl. (2) 17, 277–364.

In the articles of this series Kolmogorov put the question with full clarity: how is the *smoothness of the class of functions* related to the *rapidity of its approximation*? In this series he was concerned with ε -entropy, that is, with approximation at finitely many points, but it is completely clear that it is possible to pose the question about approximation by finite-dimensional spaces, Fourier methods, and so on.

Kolmogorov himself obtained the following result in the above-mentioned work of 1956. Let $H_\infty^\alpha(I^n)$ denote the class of functions $x(\cdot)$, defined on the n -dimensional cube, with smoothness α , understood in the following sense. Let $\alpha = r + \beta$, where r is an integer, $r \geq 0$, and $0 < \beta \leq 1$. Then

$$H_\infty^\alpha(I^n) = \left\{ x(\cdot) \mid x(\cdot) \in C^r(I^n), \left| \frac{\partial^r x(t)}{\partial t_1^{r_1} \dots \partial t_n^{r_n}} - \frac{\partial^r x(t')}{\partial t_1^{r_1} \dots \partial t_n^{r_n}} \right| \leq \right. \\ \left. \leq |t - t'|^\beta \text{ for all } (r_1, \dots, r_n), \sum_{i=1}^n r_i = r \right\}.$$

Classes of this kind are now called *Nikol'skii classes*.

Theorem 6. For any $n \in \mathbb{N}$ and $\alpha > 0$, we have the asymptotic behaviour

$$\mathcal{E}_\varepsilon(BC(I^n) \cap H_\infty^\alpha(I^n), C(I^n)) \asymp \varepsilon^{-n/\alpha}, \quad \varepsilon \rightarrow 0.$$

As well as smooth functions, analytic functions were also considered. Let G be a domain in $\mathbb{C}^n$ containing the cube I^n . Let A^G denote the collection of functions $x(\cdot)$, holomorphic in G , for which $|x(z)| \leq 1$ for all $z \in G$. Kolmogorov proved the following result:

Theorem 6'. $\mathcal{E}_\varepsilon(A^G, C(I^n)) \asymp (\log 1/\varepsilon)^{n+1}$.

From comparison of Theorems 6 and 6', a programme of research stands out particularly clearly, linked by the words "approximation" and "smoothness". How are they related to each other? How much more massive is the class of smooth functions compared with analytic functions? And so on.

With this we conclude our survey of Kolmogorov's work in approximation theory. In §7 we shall touch on the last paper, [3] No. 7, which has not yet found a fitting continuation.

But now we shall see how all these ideas and results of Kolmogorov have been developed.

We shall proceed in the same order: exact bounds for approximations to classes, diameters, inequalities for derivatives, diameters of finite-dimensional sets, the relation between smoothness and speed of approximation. In the closing section we shall give proofs of some theorems, so that the reader can get an idea of the *methods* used in this branch of mathematical analysis.

§2. Exact estimates for approximations to classes

Kolmogorov's work on bounds for the remainders of Fourier series were soon developed in the researches of Favard. In 1936–1937 Favard solved the problem of finding the *deviation of the class $W_{\infty}^r(\mathbf{T})$ from the space $\mathcal{T}_{n-1}$ of trigonometric polynomials in the metric $C(\mathbf{T})$* . More precisely, he obtained the following formula ($n, r \in \mathbf{N}$):

$$(1) \quad d(W_{\infty}^r(\mathbf{T}), \mathcal{T}_{n-1}, C(\mathbf{T})) := \sup_{x(\cdot) \in W_{\infty}^r(\mathbf{T})} \inf_{y(\cdot) \in \mathcal{T}_{n-1}} \|x(\cdot) - y(\cdot)\|_{C(\mathbf{T})} = K_r n^{-r},$$

where the K_r are the constants we have already met in §1.3, there called the *Favard constants* (although they had already occurred in the work of Euler). After this, a number of similar exact formulae for the approximation of classes of smooth or analytic functions by trigonometric polynomials were obtained by the Soviet mathematicians N.I. Akhiezer and M.G. Krein, who significantly extended Favard's method.

Favard's approximation method, leading to (1), is a linear (though rather complicatedly constructed) method of summation of a Fourier series.

The articles of Kolmogorov [1] No. 27, Favard [4], and Akhiezer and Krein [5] laid the foundations of a new trend, in which they investigated extremal problems in approximation theory, consisting in the discovery of exact upper bounds of approximations by some definite method to classes of functions.

Kolmogorov's pupil S.M. Nikol'skii played a particularly important role in the development of this area. An excellent account of Nikol'skii's role in approximation theory can be found in the article [18] by Korneichuk, dedicated to him on his 80th birthday. In particular, he says there that in the period 1940–1951 Nikol'skii published a series of papers which "radically changed and enlarged the frontier of research on approximation theory in our country, and gave a powerful impetus to the development of this area of mathematics, whose effect has not weakened to the present day ... One can point out a whole series of leading trends in the modern theory of approximation that have been formed and intensively developed on the basis and under the influence of the ideas and methods of Nikol'skii: linear methods of summation of Fourier series and interpolation polynomials, duality of extremal problems and precise constants of best approximation, optimization of methods of approximate integration, approximation characteristics of classes of functions defined by a majorant of the modulus of continuity, approximation of functions on a finite interval by polynomials". The last of these is a natural continuation of researches in the sphere of ideas of Bernstein and Jackson, but all the rest follow in one way or another the path laid down by Kolmogorov's article [1] No. 27, and the researches of Favard, Akhiezer, and Krein.

Kolmogorov's ideas on approximation to classes and exact constants, significantly developed and enriched by the work of Nikol'skii, have led to the formation of several mathematical schools. In Moscow the activity of Stechkin played an important part in developing this area, and he, it seems to me, must be counted among the students of Kolmogorov in this area of mathematics. The centre of attraction of the Moscow school was in the Steklov Mathematical Institute. Here we must mention A.V. Efimov, K.I. Oskolkov, V.N. Temlyakov, S.A. Telyakovskii, and others. The school of V.K. Dzyadyk (V.N. Konovalov, A.I. Stepanets, and others) was formed at the Kiev Mathematical Institute, and at Dnepropetrovsk the school of N.P. Korneichuk⁽¹⁾ (A.A. Ligun, V.P. Motornyi, and others). Both Dzyadyk and Korneichuk are students of Nikol'skii.

The school of Stechkin (V.V. Arestov, Yu.N. Subbotin, N.I. Chernykh, and others) works intensively at Sverdlovsk. A.F. Timan has done much for the development of this area (he too is a pupil of Nikol'skii). Approximation theory is actively developing in Alma-Ata (where Korneichuk's student A.A. Zhensykbayev works), in Yaroslavl' (where Timan's student Yu.A. Brudnyi works), and in many other towns.

There are now very many survey articles and monographs dedicated to this subject. There is, above all, the work of Nikol'skii: his 1945 article [6], his book on quadrature formulae [7], monographs by Akhiezer [8], Dzyadyk [9], Korneichuk [10], [11], [12], Korneichuk, Ligun and Dorenin [13], Stepanets [14], [15], Timan [16], Tikhomirov [17], and the article of Korneichuk [18] mentioned above (see also the commentaries of Telyakovskii and Tikhomirov in [1]). This allows me not to expound in a detailed succession all the results that have been obtained in this area. I have confined myself to stating theorems which form in some respect "boundary points" in the development of the various branches that separate from the main trunk.

2.1. Generalization and development of Kolmogorov's result on the remainders of Fourier series.

Let X be a normed space, $C \subset X$ a subspace, and Λ_n a family of linear operators from X to X . The study of the asymptotic behaviour with respect to n of the quantity $\lambda(C, \Lambda_n, X) = \sup \{ \|x - \Lambda_n x\| \mid x \in C \}$ is sometimes called the *Kolmogorov-Nikol'skii problem*. In this notation the quantities $\varphi_{n,r}$, introduced in §1.1, in connection with Kolmogorov's paper [1] No. 27, are none other than $\lambda(W_\infty^r(\mathbf{T}), S_n, C(\mathbf{T}))$.

A more complete result, immediately related to estimates for $\varphi_{n,r}$, was obtained by Stechkin [19].

⁽¹⁾Now working in the Mathematical Institute of the USSR Academy of Sciences.

Theorem 1. Let $n \in \mathbf{N}$, $r \in \mathbf{R}$, $r \geq 1$. Then

$$\varphi_{nr} = n^{-r} (8\pi^{-2} \mathbf{K}(\exp(-r/n)) + O(r^{-1}))$$

uniformly in n and r , where $\mathbf{K}(x) := \int_0^{\pi/2} (1 - x^2 \sin^2 t)^{-1/2} dt$, $0 \leq x < 1$, is the complete elliptic integral of the first kind.

A similar formula holds in the space $L_1(\mathbf{T})$. Here $r \in \mathbf{R}_+$, and the derivative is understood in the sense of Weyl.

This result was preceded by work of Pinkevich, Nikol'skii, Efimov, Telyakovskii, Sokolov, Selivanov, and Natanson (see the articles cited in [19]).

Nicol'skii introduced classes of functions into approximation theory, which were defined by moduli of continuity; these include the Nicol'skii classes mentioned in §1.6, as well as the following classes:

$$W_{\infty}^r H^{\omega}(\mathbf{T}) := \{x(\cdot) \mid x(\cdot) \in C^r(\mathbf{T}), \mid x^{(r)}(t) - x^{(r)}(t') \mid \leq \omega(\mid t - t' \mid)\},$$

where the r -th derivative is understood in the sense of Weyl. Nicol'skii proved the following fundamental result (see [14]).

Theorem 2. Let $r > 0$ and let $\omega(\cdot)$ be a convex upwards modulus of continuity. Then

$$\begin{aligned} \varphi_{nr\omega} &= \lambda(W_{\infty}^r H^{\omega}(\mathbf{T}), S_n, C(\mathbf{T})) = \\ &= 2\pi^{-2} n^{-r} \log n \int_0^{\pi/2} \omega\left(\frac{4t}{2n+t}\right) \sin t \, dt + O(n^{-r}\omega(n^{-1})). \end{aligned}$$

Kolmogorov's result follows from this with $\omega(t) = t$ and r an integer.

The essential development in this area was effected by Efimov and Stepanets (see [14]).

Approximation to functions of several variables by rectangular partial sums of Fourier series has been investigated by Bugaets, Stepanets, and others. Many authors have studied the Kolmogorov-Nicol'skii problem for other operators: Fejér, de la Vallée Poussin, Jackson, Rogosinski, and others (see [9]–[17]).

2.2. Generalization and development of the results of Favard and Akhiezer-Krein.

It must be said that the method of approximation leading to formula (1) is *linear*, so that the theorem of Favard-Akhiezer-Krein can itself be regarded as a "Kolmogorov-Nicol'skii problem". However, it is not this which makes us touch on this theme, but its importance for what follows and for the connection with diameters.

The definitive result on Favard's problem was far from immediate, but was the outcome of efforts lasting many years by Dzyadyk, Stechkin, and Sun-Yun Shen. The fullest result was obtained by Dzyadyk in 1974, though

his first article on the subject was written in 1953 (on this see [18]). We shall give a special case of these results that immediately generalizes (1).

Theorem 3. Let $n \in \mathbf{N}$, $r > 0$; then

$$d(W_\infty^r(\mathbf{T}), \mathcal{F}_n, C(\mathbf{T})) = n^{-r} K_r,$$

where

$$K_r = 4\pi^{-1} (\sin r\pi/2) \sum_{k \in \mathbf{N}} (2k-1)^{-(r+1)} \text{ for } 0 < r \leq 1$$

and

$$K_r = 4\pi^{-1} \left(\sum_{k \in \mathbf{N}} (2k-1)^{-(r+1)} \sin((2k-1)\pi\theta - r\pi/2) \right),$$

$0 < \theta \leq 1$ and θ is a root of the equation

$$\sum_{k \in \mathbf{N}} (2k-1)^{-r} \cos((2k+1)\theta\pi - r\pi/2) = 0.$$

For $r \in \mathbf{N}$ we get (1).

The Favard problem for the classes $W_\infty^r H^\omega(\mathbf{T})$, $r \in \mathbf{Z}_+$, was solved by Korneichuk for convex moduli of continuity (see [10]–[12]).

Theorem 4. Let $\omega(\cdot)$ be a convex (upwards) modulus of continuity, $r \in \mathbf{Z}_+$. Then

$$d(W_\infty^r H^\omega(\mathbf{T}), \mathcal{F}_n, C(\mathbf{T})) = \|\tilde{x}_{nr\omega}(\cdot)\|_{C(\mathbf{T})},$$

where the functions $\tilde{x}_{nr\omega}(\cdot)$ with period $2\pi/n$ are defined inductively:

$$\tilde{x}_{n0\omega}(t) = \begin{cases} -1/2\omega(\pi/n - 2t), & 0 \leq t \leq \pi/2n, \\ 1/2\omega(2t - \pi/n), & \pi/2n \leq t \leq \pi/n, \end{cases}$$

and extended as even functions to $[-\pi/n, \pi/n]$ and then by periodicity:

$$\tilde{x}_{n2k-1\omega}(t) = \int_0^t \tilde{x}_{n2k-2\omega}(\tau) d\tau, \quad \tilde{x}_{n2k\omega}(t) = \int_{\pi/2n}^t \tilde{x}_{n2k-1\omega}(\tau) d\tau.$$

With $\omega(t) = t$ we again obtain (1).

There are many articles on Favard's problem in other metrics, in particular in $L_1(\mathbf{T})$ (see [12]).

2.3. The Kolmogorov-Nikol'skii and Favard problems for analytic functions on the circle.

With a view to the comparison of smooth and analytic functions we shall give some results similar to Theorems 1 of §1.1 and 1 of §2.1 in the case when $\mathbf{T}$ is replaced by the unit disc $D = \{z \in \mathbf{C} \mid |z| \leq 1\}$, the classes $W_p^r(\mathbf{T})$ by the classes $\mathcal{H}_p^r(D)$, the set of functions $x(\cdot)$ analytic in $\text{int } D$, for which

$$\int_{\mathbf{T}} |x^{(r)}(\rho e^{it})|^p dt \leq 1 \text{ for all } \rho, \quad 0 < \rho < 1, \quad 1 \leq p < \infty$$

or

$$|x^{(r)}(z)| \leq 1 \text{ for all } z, |z| < 1, p = \infty,$$

the Fourier sums are replaced by the Taylor sums

$$T_n(x(z)) = \sum_{k \in \mathbf{N}} x_{k-1} z^{k-1} \rightarrow T_n x(z) = \sum_{k=0}^n x_k z^k,$$

and the space of trigonometric polynomials by the space $\mathcal{F}_n = \text{lin} \{z^k, 0 \leq k \leq n\}$ of algebraic polynomials of degree $\leq n$.

A problem analogous to that of Kolmogorov in §1.1 has been solved by Stechkin [20] for the classes $\mathcal{H}_\infty^r(D)$.

Theorem 5. *Let r be a natural number. Then*

$$\tau_{n,r} = \sup \{ \|x(\cdot) - T_n x(\cdot)\|_{C(D)} \mid x(\cdot) \in \mathcal{H}_\infty^r(\mathbf{T}) \} = \pi^{-1} n^{-r} \log n + O(n^{-r}).$$

The case $r = 0$ had been investigated earlier by Landau in 1916.

A problem analogous to Favard's ($p = \infty$) was solved by Babenko [21], and his result was generalized by Taikov [22] for $1 \leq p < \infty$.

Theorem 6. *Let r and n be natural numbers, $1 \leq p \leq \infty$. Then*

$$d(\mathcal{H}_p^r(D), \mathcal{P}_{n-1}, L_p(\partial D)) = (n(n-1) \dots (n-r+1))^{-1}.$$

§3. Diameters

In this section we shall be mainly concerned with the development of Kolmogorov's work [1] No. 28, that is, with *exact* values of the diameters of classes of functions. Other problems concerned with diameters will be discussed in §§4 and 6. The first part of this section is concerned with the definitions of various diameters.

3.1. Definitions of diameters.

The size of the Kolmogorov diameter answers the question: how closely can a given class of functions be approximated by approximations from a subspace of given dimension? Classically the approximations were from the space $\mathcal{P}_n$ of algebraic polynomials, or $\mathcal{T}_n$ of trigonometric polynomials of degree $\leq n$. This naturally raises the question: for a given class of functions, can we not find a subspace of the same dimension as $\mathcal{P}_n$ or $\mathcal{T}_n$ giving better approximations? But in classical approximation theory questions of another kind were also considered. For example the characteristics of approximation by Fourier sums were investigated in [1] No. 27. We may ask what subspace will have the best properties for approximation to a given class by the Fourier method. Other linear methods of approximation have also been investigated—Fejér sums, sums of de la Vallée Poussin, Jackson, and others. Again we may ask what *linear* (finite-dimensional) method approximates a given class best of all. Finally, non-linear approximations have also been

studied classically, for example, rational approximations. And besides we may ask whether there is not some other set with the same number of parameters that approximates the given class better?

All these questions lead us to introduce corresponding quantities—Fourier diameters, linear and Aleksandrov diameters. Let us give their definitions.

Let X be a normed space, C and A subsets of X . Let $\text{Lin}_n(X)$ denote the collection of all n -dimensional subspaces of X , and $\mathcal{E}(C, A)$ and $\mathcal{C}(C, A)$ denote respectively the set of all mappings and all continuous mappings of C into A . Further let $\mathfrak{A}$ be a family $\{A\}$ of approximating sets and $\mathcal{F} = \mathcal{F}(\mathfrak{A})$ a subset of mappings $\{F\}$ of C into A . We define the quantity

$$(1) \quad p_{(\mathfrak{A}, \mathcal{F})}(C, X) = \inf_{(A, F) \in (\mathfrak{A}, \mathcal{F})} \sup_{x \in C} \|x - F(x)\|.$$

With different interpretations of the pairs $(\mathfrak{A}, \mathcal{F})$, we obtain all the diameters we have mentioned.

a) Let H be a Hilbert space densely embedded in X , and let $C \subset H$, $\mathfrak{A}_1 = \text{Lin}_n(H)$, $\mathcal{F}_1 = \bigcup_{L_n \in \mathfrak{A}_1} PL_n$, where PL_n is the orthogonal projection

onto L_n . Then $p_{(\mathfrak{A}_1, \mathcal{F}_1)}(C, X) = \varphi_n(C, X)$ is the *Fourier diameter* of C in X , characterizing the best approximation by the Fourier method (Temlyakov, 1980).

b) If $\mathfrak{A}_2 = \text{Lin}_n(X)$, and $\mathcal{F}_2 = \bigcup_{L_n \in \mathfrak{A}_2} \mathcal{L}(X, L_n)$ (as usual $\mathcal{L}(X, Y)$ is the set of continuous linear operators from X to Y), then $p_{(\mathfrak{A}_2, \mathcal{F}_2)}(C, X) = \lambda_n(C, X)$ is the *linear diameter*, characterizing the best n -dimensional linear method (Tikhomirov, 1960).

c) If $\mathfrak{A}_3 = \text{Lin}_n(X)$, and $\mathcal{F}_3 = \bigcup_{L_n \in \mathfrak{A}_3} \mathcal{E}(X, L_n)$, then, as it is easy to see, $p_{(\mathfrak{A}_3, \mathcal{F}_3)}(C, X) = d_n(C, X)$ is the Kolmogorov diameter (Kolmogorov, 1936).

d) If $\mathfrak{A}_4$ is the collection of n -dimensional compacta⁽¹⁾ K_n in X , and $\mathcal{F}_4 = \bigcup_{K_n \in \mathfrak{A}_4} \mathcal{C}(C, K_n)$, then $p_{(\mathfrak{A}_4, \mathcal{F}_4)}(C, X) = a_n(C, X)$ is the Aleksandrov diameter of C in X , characterizing finite-dimensional continuous operators from the point of view of their approximating properties (Aleksandrov, 1933).

3.2. Splines as a best approximation method.

The article [1] No. 28 long remained unnoticed. It was not until 1954 that Stechkin found the weak asymptotics of the diameters of the classes $W_1^r(\mathbf{T})$ in $L_2(\mathbf{T})$ and $W_\infty^r(\mathbf{T})$ in $C(\mathbf{T})$. The first exact result after Kolmogorov's was obtained by the author in 1960 (and refined for even $m = 2n$ in 1969).

⁽¹⁾By the well-known theorem of Aleksandrov, we can restrict ourselves to n -dimensional complexes.

Theorem 1 [17]. *Let $r, n \in \mathbf{N}$. Then*

$$(1) \quad d_{2n-1}(W_{\infty}^r(\mathbf{T}), C(\mathbf{T})) = d_{2n}(W_{\infty}^{r+1}(\mathbf{T}), C(\mathbf{T})) = n^{-r}K_r,$$

where K_r are the Favard constants; $d_0 = \infty$.

The other diameters all have the same values:

$$\varphi_{2n-1} = \varphi_{2n} = \lambda_{2n-1} = \lambda_{2n} = a_{2n-1} = a_{2n} = n^{-r}K_r.$$

There are very many mathematicians at present working on extremal problems in the theory of diameters. The group working in Moscow is centred on Moscow University (Tikhomirov and his students Buslaev, Din' Zung, and others). Babenko and Erokhin have played a very large role in the development of this area, which has also been studied intensively abroad (Micchelli, Pinkus, Fisher, Höllig, and others). We refer to the monographs [10]–[13], [17], [26] for the results obtained.

If we compare Theorem 1 with Favard's theorem (see (1) in §2), we see that the space $\mathcal{T}_n$ is extremal also for the class $W_{\infty}^r(\mathbf{T})$ (if it is approximated in the metric $C(\mathbf{T})$). However, it soon turns out that the space $\mathcal{T}_n$ is not unique. There exist other extremal subspaces and, in a certain sense, more natural optimal methods of approximation than Favard's.

These extremals include the subspace of *splines*.

Let $S_{2n,m}$, $m \in \mathbf{N}$, denote the set of functions $y(\cdot)$ from $C^{m-1}(\mathbf{T})$ such that $y^{(m)}(\cdot)$ is a piecewise-constant function, constant on the intervals $[k\pi/n, (k+1)\pi/n]$, $k = -n, \dots, n-1$. This is a $2n$ -dimensional space, consisting of continuous functions if $m \geq 2$. In 1969 (and in special cases already in 1960) the author discovered that

$$(1') \quad d(W_{\infty}^r(\mathbf{T}), S_{2n,r-1}, C(\mathbf{T})) = n^{-r}K_r.$$

In consequence of Theorem 1 this means that the space $S_{2n,r-1}$ is extremal. Here the best method of approximation is realised by interpolation of $x(\cdot)$ at equidistant points. Later, using this result, Ligun [23] obtained the following interesting generalization.

Theorem 2. *Let r be a natural number and $m \geq r-1$. Then*

$$d(W_{\infty}^r(\mathbf{T}), S_{2n,m}, C(\mathbf{T})) = n^{-r}K_r.$$

Thus the whole family of subspaces $\{S_{2n,m}\}_{n \in \mathbf{N}}$, $m \geq r-1$, turns out to be extremal. "Passage to the limit" as $m \rightarrow \infty$ is possible in principle, and then Favard's theorem becomes a "special case" of the spline theorem (Velikin [24]).

It is natural to ask what happens in the non-periodic case for the pair $(W_{\infty}^r(I), C(I))$, where

$$W_{\infty}^r(I) = \{x(\cdot) \in C^{r-1}(I) \mid x^{(r-1)}(\cdot) \in AC(I), \|x^{(r)}(\cdot)\|_{L_{\infty}(I)} \leq 1\}.$$

In 1969 the author obtained the following result.

Theorem 3. a) For any $n, r \in \mathbf{N}$, $n \geq r$, there is a unique function $x_{nr}(\cdot) \in W_{\infty}^r(I)$ such that $|x_{nr}^{(r)}(\cdot)| \equiv 1$, $x_{nr}(\cdot)$ has n -alternance, and the number of sign changes of $x_{nr}^{(r)}(\cdot)$ is $n-r$.

$$\text{b)} \quad d_n(W_{\infty}^r(I), C(I)) = \|x_{nr}(\cdot)\|_{C(I)} = \lambda_{nr}^{-1}.$$

The other diameters, φ_n , λ_n , and a_n , also have the same values.

The functions $x_{nr}(\cdot)$ are splines; they have been called *Chebyshev splines* ($x_{rr}(\cdot)$ is proportional to a Chebyshev polynomial). Since the Chebyshev polynomials have remarkable extremal properties, the Chebyshev splines have many of these properties as well. We shall have occasion to discuss this in §4.

3.3. Spectra and diameters.

In the middle of the 1960's the author thought of a plan for solving the diameter problem for the pair $(W_p^r(I), L_q(I))$ for $p \geq q$. The idea was that the diameters are "spectral numbers" of some non-linear differential operator.

To explain the essence of the matter, we return to Theorem 2 of §1.2. This theorem of Kolmogorov's states that $d_n(W_2^r(I), L_2(I)) = \lambda_n^{-1}$, where λ is an eigenvalue of the equation $x^{(2r)} + (-1)^{r+1}\lambda^2 x = 0$ with boundary conditions $x^{(r+k)}(0) = x^{(r+k)}(1)$, $r \leq k \leq 2r-1$. This differential equation is the *Euler equation* for the following isoperimetric problem (concerning extremal problems here and later see [25]):

$$(2) \quad \int_I x^2 dt \rightarrow \sup, \quad \int_I (x^{(r)})^2 dt \leq 1.$$

Here the boundary conditions are nothing but *transversality conditions*. (Problem (2) of course has no solution; since its value is infinite we are concerned with conditions for a stationary value.) So the conjecture arises that the diameters $d_n(W_p^r(I), L_q(I))$ for $p \geq q$ can be expressed in terms of the stationary values of the isoperimetric problem

$$(2') \quad \int_I |x|^q dt \rightarrow \sup, \quad \int_I |x^{(r)}|^n dt \leq 1.$$

Hence there arises a canonical system of differential equations

$$(3) \quad x^{(r)} = y_{(p)}, \quad y^{(r)} = (-1)^r \lambda^q x_{(q)} = 0, \quad y^{(i)}(0) = y^{(i)}(1) = 0, \quad 0 \leq i \leq r-1,$$

equivalent to the equation

$$(3') \quad ((x^{(r)})_{(p)})^{(r)} + (-1)^{r+1}\lambda^q x_{(q)} = 0, \quad ((x^{(r)})_{(p)})^{(i)}(0) = ((x^{(r)})_{(p)})^{(i)}(1) = 0,$$

$0 \leq i \leq r-1$, where for a number a , $a_{(s)}$ denotes the number $|a|^{s-1} \operatorname{sign} a$. With $p = q = 2$, from (3') we obtain the linear equation in Kolmogorov's paper [1] No. 28.

Theorem 4 (existence of the spectrum). a) For any $n, r \in \mathbf{N}$, $n \geq r$, and q with $1 < q < \infty$, there is a function $x_{nr\infty q}(\cdot) \in W_{\infty}^r(I)$ and a number $\lambda_{nr\infty q}$ such that $|x_{nr\infty q}^{(r)}(t)| \equiv 1$, the number of sign changes of $x_{nr\infty q}^{(r)}(\cdot)$ (equal to the number of its simple zeros) is $n - r$, and system (3) is satisfied, that is, the function $y_{nr\infty q}(\cdot)$ defined by the relations $y_{nr\infty q}^{(r)}(\cdot) = (-1)^r \lambda_{nr\infty q}^q (x_{nr\infty q}(\cdot))^q$, $y_{nr\infty q}^{(i)}(0) = 0$, $0 \leq i \leq r - 1$, satisfies the equations $x_{nr\infty q}^{(r)}(\cdot) = \text{sign } y_{nr\infty q}(\cdot)$, $y_{nr\infty q}^{(i)}(1) = 0$, $0 \leq i \leq r - 1$.

b) For any $n, r \in \mathbf{N}$, $n \geq r$, and p and q with $1 < q \leq p < \infty$, there is a function $x_{nrpq}(\cdot) \in W_{\infty}^r(I)$ and a number λ_{nrpq} such that the number of sign changes of $x_{nrpq}^{(r)}(\cdot)$ (equal to the number of its simple zeros) is $n - r$, and equations (3') are satisfied.

Here the following equality holds:

$$(4) \quad d_n(W_p^r(I), L_q(I)) = \lambda_{nrpq}^{-1}.$$

There is a similar result in the periodic case, where for $p = \infty$ the "spectral functions" $\tilde{x}_{nr\infty q}(\cdot)$ are none other than the Euler functions of the argument nt : $\tilde{x}_{nr\infty q}(t) = x_r(nt)$ (the functions $x_r(\cdot)$ were described in §1.3).

Formula (4) has a very long history. The cases $p = q = \infty$ and $p = q = 2$ have already been mentioned. The case $p = q$ was obtained by Tikhomirov and Babadzhanyan in 1967 (for $r = 1$), and in the general case by Pinkus in 1983. The case $p = \infty$ and its dual $q = 1$ in the periodic case was obtained independently by Ligun, Makovoz, and Pinkus in 1979, in the non-periodic case by Tikhomirov, Micchelli, and Pinkus, and in the case $p \geq q$ with $r = 1$ by Tikhomirov (see [17], [26]–[29]).

The search for exact values of the Kolmogorov diameters in the "lower triangle", when $p < q$, is a difficult problem. The only result obtained so far (Ismagilov; see [17]) is the following.

Theorem 5. Let $n \in \mathbf{N}$, $r > 1/2$. Then

$$d_{2n-1}(W_1^r(\mathbf{T}), L_2(\mathbf{T})) = d_{2n}(W_1^r(\mathbf{T}), L_2(\mathbf{T})) = \left(\sum_{k \geq n} k^{-2r} \right)^{1/2}.$$

3.4. Diameters of analytic functions.

We only give one result here.

Theorem 6. Let $r \in \mathbf{N}$, $n > r$, $1 \leq p \leq \infty$. Then:

$$\begin{aligned} \text{a)} \quad & d_n(\mathcal{E}_p^r(D_R), L_p(\partial D)) = \frac{(n-r)!}{n!R^n}; \\ \text{b)} \quad & d_n(\mathcal{E}_p^r(D_R), L_p(D)) = \frac{(n-r)!}{n!R^{n-r}(np+2)^{1/p}}. \end{aligned}$$

Here $D_R = \{z \in \mathbf{C} \mid |z| \leq R\}$, $R < 1$, and the class $\mathcal{E}_p^r$ was defined in the previous section. Theorem 6 a) was proved by Babenko, Tikhomirov, and Taikov. Babenko proved the upper bound for $p = \infty$, and Taikov extended it to the case $1 \leq p < \infty$. In 1960 Tikhomirov worked out a general method for lower bounds, which he applied for $p = \infty$. Taikov showed that

this method also worked for $1 \leq p < \infty$ (see [22]). Theorem 6 b) was proved by Pinkus [27].

§4. Inequalities for derivatives

Here we shall discuss developments from Kolmogorov's paper [1] No. 40. It is well known that the first results concerned with finding exact inequalities for derivatives of the form

$$(1) \quad \|x^{(k)}(\cdot)\|_{L_q(I)} \leq K \|x(\cdot)\|_{L_p(I)}^\alpha \|x^{(n)}(\cdot)\|_{L_r(I)}^\beta$$

(where $0 \leq k < n$ are integers, $1 \leq p, q, r \leq \infty$, $I = \mathbf{R}$ or $\mathbf{R}_+$), were obtained by Landau (1913) and Hadamard (1914) ($k = 1$, $n = 2$, $p = q = r = \infty$, I respectively $\mathbf{R}_+$ or $\mathbf{R}$). Before Kolmogorov's paper Hardy, Littlewood, and Pólya had investigated the case $p = q = r = 2$, where $n \in \mathbf{N}$, $0 < k < n$. Kolmogorov's paper stimulated a general formulation of the problem of finding best-possible constants in (1) for fixed values of the parameters (p, q, r, k, n) on the line or half-line (here the numbers α and β are uniquely determined by the formulae $\alpha = \frac{n-k-r+q^{-1}}{n-r^{-1}+p^{-1}}$, $\beta = 1 - \alpha$).

There are very few results similar to Kolmogorov's, and it is easy to list them all.

Theorem 1. a) In the case $I = \mathbf{R}$, the exact constant $K = K(p, q, r, k, n)$ for any $n \in \mathbf{N}$ and integers k , $0 \leq k < n$, has been found (for fixed p, q, r) in the following four cases:

α) $p = q = r = \infty$ (Kolmogorov, 1939); β) $p = q = r = 2$ (Hardy, Littlewood, and Pólya, 1934); γ) $p = q = r = 1$ (Stein, 1957); δ) $p = r = 2$, $q = \infty$ (Taikov, 1968).

b) In the case $I = \mathbf{R}_+$ the exact constant (for any $n \in \mathbf{N}$ and $0 \leq k < n$, k an integer) has been found explicitly (for fixed p, q, r) in two cases:

α) $p = q = r = 2$ (Lyubich, 1960; Kuptsov, 1975), β) $p = r = 2$, $q = \infty$ (Gabushin, 1969) (see the commentary by Magaril-Il'yaev and Tikhomirov in [1]).

Interest in this area grew after Stechkin [31] revealed the close connection between the problem of computing the exact constant in (1) and approximation theory (in the Kolmogorov sense of "approximation to classes"), namely the problem of best-possible approximation to the differentiation operator on classes of functions. Then a series of 20 partial solutions was obtained, similar to the Landau and Hadamard cases, when all the values of the parameters were fixed. Most of these solutions were obtained by students of Stechkin and Tikhomirov (Arestov, Buslaev, Berdyshev, Gabushin, Magaril-Il'yaev, and others).

Older work by Nagy and Matorin, as well as Kwong and Zettl, Konovalov, and others is concerned with the same range of problems. (For references see the above-mentioned commentary by Magaril-Il'yaev and Tikhomirov.)

Closely related to this topic is the problem of intermediate derivatives on an interval, whose origins go back to the beginnings of the development of approximation theory (Chebyshev, Markov, Zolotarev). These problems have been studied by Tikhomirov, Pinkus, Burenkov, Zvyagintsev, and others. We shall give one result, related to Theorem 3 of §3.

Theorem 2. a) For $n, r \in \mathbf{N}$, $n \geq r$, and a point β , not coinciding with a zero of the Chebyshev spline $x_{nr\infty\infty}(\cdot)$, there is a unique function $x_\beta(\cdot) = x_{nr\beta\infty\infty}(\cdot)$ such that $|x_\beta^{(r)}(t)| \equiv 1$, the number of sign changes of $x_\beta^{(r)}(\cdot)$ is $n-r$, the function has $n-1$ alternance and satisfies $x_\beta(\beta) = 0$. Also for any $\gamma > 0$

$$\|x_{n-1r\infty\infty}(\cdot)\| \geq \gamma > \|x_{nr\infty\infty}(\cdot)\|$$

there is a unique number β , $\beta_{nr} \leq \beta < \infty$, such that

$$\|x_{nr\beta\infty\infty}(\cdot)\|_{C(I)} = \gamma.$$

b) For any point $\tau \in I$ and any function $x(\cdot) \in W_\infty^r(I)$, the exact inequality $|x^{(h)}(\tau)| \leq |x_{nr\beta\infty\infty}^{(h)}(\tau)|$ is satisfied, where β is chosen so that $\|x_{nr\beta\infty\infty}(\cdot)\|_{C(I)} = \|x(\cdot)\|_{C(I)}$.

c) The following exact inequality holds for intermediate derivatives:

$$\|x^{(h)}(\cdot)\|_{C(I)} \leq \|x_{nr\beta\infty\infty}^{(h)}\|_{C(I)},$$

where β is again chosen so that $\|x_{nr\beta\infty\infty}(\cdot)\|_{C(I)} = \|x(\cdot)\|_{C(I)}$.

The functions $x_{nr\beta\infty\infty}(\cdot)$ have been called *Zolotarev splines* (they generalize the classical Zolotarev polynomials and inherit their extremal properties).

The results a) were obtained by Tikhomirov [17] and Pinkus, and this concludes the topic of inequalities for derivatives going back to [32]. Theorem b) is due to Zvyagintsev. Pinkus also solved the problem of "intermediate derivatives at an interior point of an interval" (for polynomials this was solved by Markov). Here new "Pinkus" splines appear.

§5. Diameters of finite-dimensional sets

The Kolmogorov-Petrov-Smirnov-Mal'tsev-Stechkin theorem, which we spoke of in §1.4, has also been considerably developed. First we shall state the exact results.

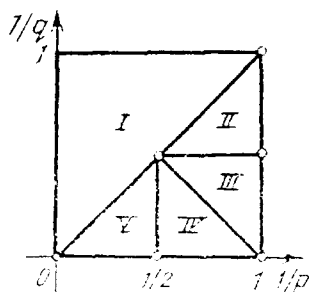
Let us recall that l_p^n denotes the Banach space $\mathbf{R}^n$ with the norm

$\|x\|_{l_p^n} = (\sum_{i=1}^n |x_i|^p)^{1/p}$, $1 \leq p < \infty$, $\|x\|_{l_\infty^n} = \max_{1 \leq i \leq n} |x_i|$, and Bl_p^n is the unit ball in l_p^n .

Theorem 1. For any non-negative integer k , natural number n , and any $p, q, 1 \leq p, q \leq \infty$, we have

$$a_k(Bl_p^n, l_q^n) = \begin{cases} (n-k)^{1/q-1/p}, & p \geq q, \\ (k+1)^{1/q-1/p}, & p \leq q. \end{cases}$$

For $p \leq q$ this result was obtained by Stesin (see [17]), and for $p \geq q$ by Khodulev.



Theorem 2. For any non-negative integer k , natural number n , and p, q satisfying $1 \leq q \leq p \leq \infty$, we have

$$\lambda_k(Bl_p^n, l_q^n) = d_k(Bl_p^n, l_q^n) = (n-k)^{1/q-1/p}.$$

This result was obtained independently by Pich and Stesin (see [17]).

At the end of the 1970's it became clear that problems on the diameters of functional classes could be reduced to the calculation of the diameters of the pairs (Bl_p^n, l_q^n) . Many people studied this problem, including Tikhomirov and Solomyak, Ismagilov, Gluskin, and others). The first qualitatively new result here was obtained by Ismagilov [32]. Bounds necessary for obtaining the asymptotics of the Kolmogorov diameters $d_n(W_p^r(\mathbf{T}), L_q(\mathbf{T}))$ for any p and q were first obtained by Kashin [33]. Subsequently Gluskin obtained substantial refinements of this result.

Let us put (see the figure)

$$\Phi(k, n, p, q) = \begin{cases} \max(k^{1/q-1/p}, (1-n/k)^{(1/q-1/q)/(1/p-1/2)}) & \text{in II,} \\ \max(k^{1/q-1/p}, \min(1, k^{1/q}n^{-1/2})(1-n/k)^{1/2}) & \text{in III and IV,} \\ \min(1, k^{1/q}n^{-1/2})(1/p-1/q)(1/2-1/q) & \text{in V.} \end{cases}$$

$$\Psi(k, n, p, q) = \begin{cases} \Phi(k, n, p, q), & 1 \leq p < q \leq p', \\ \Phi(k, n, q', p'), & \max(p, p') < q < \infty. \end{cases}$$

Theorem 3. Let $1 \leq p \leq q < \infty$, or $2 \leq p < \infty, q = \infty, k < n$. Then

- a) $d_k(Bl_p^n, l_q^n) \asymp \Phi(k, n, p, q);$
- b) $\lambda_k(Bl_p^n, l_q^n) \asymp \Psi(k, n, p, q).$

§6. Criteria for the closest elements

Kolmogorov's article [1] No. 49 has also given rise to a great many investigations. In Russia the topic has been developed by Garkavi, Zukhovitskii, Nikol'skii, Stechkin, Khavinson, and others. The Romanian mathematician Singer has studied it extensively, and has written several survey articles and monographs on this question. The problem studied by Kolmogorov essentially concerns convex programming, and the effects discovered reduce to the phenomenon known in convex analysis and approximation theory under the name of "refinement". The following result is in some sense the "ultimate generalization" of Kolmogorov's result.

Theorem. Let T be a compact topological space, Y a normed space, and A a convex finite-dimensional subset of $C(T, Y)$. Then for $\hat{\xi}(\cdot)$ to be the element in A that best approximates $x(\cdot)$ it is necessary and sufficient that there exist r points $t_i \in T$, where $r \leq \dim A + 1$, r linear functionals $y_i^* \in Y^*$, $\|y_i^*\| = 1$, and an element z^* belonging to the convex hull of the elements $\{\Lambda_{t_i}^* y_i^*\}_{i=1}^r$, where Λ_t^* is the operator defined by the rule $\langle \Lambda_t y^*, x(\cdot) \rangle = \langle y^*, x(t) \rangle$, such that the following relations are satisfied:

- a) $\langle z^*, x(\cdot) - \hat{\xi}(\cdot) \rangle = \|x(\cdot) - \hat{\xi}(\cdot)\|_{C(T, Y)}$;
- b) $\langle z^*, \hat{\xi}(\cdot) \rangle = sA(z^*)$,

where $sA(z^*) = \sup_{\xi(\cdot) \in A} \langle z^*, \xi(\cdot) \rangle$ is the support function of the set A .

◁ 0.1. The criterion for the element of best approximation [17], p.150.

0.2. The "refinement" theorem [17], p.43. 0.3. The Hahn-Banach

theorem. 0.4. The Banach-Alaoglu theorem. 0.5. Linear functionals attain their extremal values at boundary points.

1. Let $\hat{\xi}(\cdot)$ be an element of best approximation. Then by 0.1 there is a $z^* \in (C(T, Y))^*$, $\|z^*\| \leq 1$, such that a) and b) are satisfied. Let $f(\hat{\xi}(\cdot)) := \|z(\cdot) - \hat{\xi}(\cdot)\|_{C(T, Y)}$. From 0.1 and a) it follows that $z^* \in \partial f(\hat{\xi}(\cdot))$. Without loss of generality we may assume that $0 \in A$, so that there is a subspace $L \subset C(T, Y)$ with $A \subset L$, $\dim A = \dim L$. Let $\varphi((t, y^*), \xi(\cdot)) = \langle y^*, x(t) - \xi(t) \rangle$, $t \in T$, $y^* \in Y^*$, $\xi(\cdot) \in L$.

From 0.3 it follows that $f(\hat{\xi}(\cdot)) = \max \{\varphi((t, y^*), \hat{\xi}(\cdot)) \mid (t, y^*) \in T \times BY^*\}$ for all $\xi(\cdot) \in L$. By Theorem 0.4, BY^* is weakly compact, and hence $T \times BY^*$ is compact in the product topology. By 0.2 there is an $r \in \mathbf{N}$, $r \leq \dim L + 1$, $\{(t_i, y_i^*)\}_{i=1}^r$, $(t_i, y_i^*) \in T \times BY^*$, $i = 1, \dots, r$ (where $\|y_i^*\| = 1$

by 0.5): $-z^* \in \text{co} \bigcup_{i=1}^r \partial_{\xi(\cdot) \in \varphi((t_i, y_i^*), \hat{\xi}(\cdot))}$. Since, clearly,

$\partial_{\xi(\cdot) \in \varphi((t_i, y_i^*), \hat{\xi}(\cdot))} = \langle -\Lambda_{t_i}^* y_i^*, \hat{\xi}(\cdot) \rangle$, we have $z^* \in \text{co} \{\Lambda_{t_i}^* y_i^*, \dots, \Lambda_{t_r}^* y_r^*\}$.

2. Let $\hat{\xi}(\cdot) \in A$, and suppose that a) and b) are satisfied and $z^* \in \text{co} \{\Lambda_{i_1}^* y_{i_1}^*, \dots, \Lambda_{i_r}^* y_{i_r}^*\}$. It is clear that $\|z^*\| \leq 1 \stackrel{(i)}{\Rightarrow} \forall \xi(\cdot) \in A \Rightarrow \Rightarrow \|x(\cdot) - \hat{\xi}(\cdot)\|_{C(T, T)} \stackrel{(a)}{=} \langle z^*, x(\cdot) - \hat{\xi}(\cdot) \rangle \stackrel{\text{Id}}{=} \langle z^*, x(\cdot) \rangle - \langle z^*, \hat{\xi}(\cdot) \rangle \stackrel{(b)}{\leq} \leq \langle z^*, x(\cdot) \rangle - \langle z^*, \xi(\cdot) \rangle \stackrel{\text{Id}}{=} \langle z^*, x(\cdot) - \xi(\cdot) \rangle \stackrel{(i)}{\leq} \|x(\cdot) - \xi(\cdot)\|_{C(T, T)} \Rightarrow \hat{\xi}(\cdot)$ is an element of best approximation. $\triangleright$ Kolmogorov's criterion is a special case of this result. (This version of the theorem is due to Magaril-Il'yayev.)

§7. Smoothness and approximation

7.1. Introduction.

Beginning with the work of Bernstein and Jackson, the problem of the mutual relation between "smoothness" and "approximation" has occupied a central position in approximation theory. Both these concepts require discussion. To show the evolution of attitudes to these concepts, we shall state a number of classical theorems in which they appear in one way or another. We restrict ourselves to the one-dimensional case for the moment. We shall use notations that have been used throughout the article: T is the circle, realized as the interval $[-\pi, \pi]$ with identification of $-\pi$ and π , $d(x, A, X)$ is the distance of the point $x \in X$ from the set $A \subset X$ in the normed space X , $d(C, A, X) = \sup \{d(x, A, X) \mid x \in C\}$ is the distance of the set C from $A \subset X$, $H_p^\alpha(T)$ ($\alpha > 0$, $\alpha = r + \beta$, $r \in \mathbf{Z}_+$, $0 < \beta \leq 1$) is the Nikol'skii class consisting of the functions $x(\cdot)$ on T for which $x^{(r-1)}(\cdot) \in AC(T)$, $\int_T |x^{(r)}(t + \tau) - x^{(r)}(t)|^p dt \ll |\tau|^{p\beta}$ (with the natural modification for

$p = \infty$), where the symbol $\ll$ means that the left-hand side does not exceed $C|\tau|^\beta$ for some C independent of $t \in T$ and $x(\cdot) \in H_p^\alpha(T)$ (for definiteness we take $C = 1$); $\mathcal{T}_n$ is the set of trigonometric polynomials of degree $\leq n$.

Classical approximation theory considered problems on embedding, on approximation by polynomials, by Fourier, Fejér, or Jackson sums, on inequalities for derivatives, and so on. Here are the most significant results, due to the pioneers in each area.

The Hardy-Littlewood embedding theorem. Let $1 \leq p \leq q \leq \infty$, $\alpha > 1/p - 1/q$. Then there is a compact embedding

$$H_p^\alpha(T) \hookrightarrow H_q^{\alpha - (1/p - 1/q)}(T).$$

The Jackson-Bernstein theorem on approximation by trigonometric polynomials. If $x(\cdot) \in H_p^\alpha(T)$, then $d(x(\cdot), \mathcal{T}_n, C(T)) \ll n^{-\alpha}$, and further $d(H_p^\alpha(T), \mathcal{T}_n, C(T)) \ll n^{-\alpha}$.

Approximation of this kind is realized by the Fejér operator for $0 < \alpha \leq 1$, the Jackson operator for $0 < \alpha \leq 2$, and the de la Vallée Poussin operator for all α .

Let $x(t)$ be expanded as the Fourier series $\sum_{k \in \mathbb{Z}_+} (a_k \cos kt + b_k \sin kt)$. Let $S_n x(t)$ and $\mathcal{F}_n x(t)$ denote the quantities $\sum_{k=0}^n (a_k \cos kt + b_k \sin kt)$ and

$$\sum_{k=0}^n (1 - k/n) (a_k \cos kt + b_k \sin kt)$$

respectively. The functions $S_n x(\cdot)$ and $\mathcal{F}_n x(\cdot)$ are called *Fourier* and *Fejér sums*.

The Riesz-Lebesgue theorem on the rate of convergence of Fourier sums.

Let $x(\cdot) \in H_p^\alpha(\mathbb{T})$, $\alpha \geq 0$. Then $\|x(\cdot) - S_n x(\cdot)\|_{L_p(\mathbb{T})} \ll n^{-\alpha}$ for $1 < p < \infty$ and $n^{-\alpha} \log n$ for $p = 1, \infty$.

Fejér's theorem. If $x(\cdot) \in C(\mathbb{T})$, then $\|x(\cdot) - \mathcal{F}_n x(\cdot)\|_{C(\mathbb{T})} \rightarrow 0$. If $x(\cdot) \in H_\infty^\alpha(\mathbb{T})$, then

$$\|x(\cdot) - \mathcal{F}_n x(\cdot)\|_{C(\mathbb{T})} \ll n^{-\min(\alpha, 1)}$$

(Bernstein).

In this section we aim to observe the development of the connections which the above results establish (from the Kolmogorov point of view) when instead of $\mathcal{F}_n$ we study all the subspaces of the same dimension as $\mathcal{F}_n$, and instead of the Fourier, Fejér, Jackson, or de la Vallée Poussin sums, we consider arbitrary finite-dimensional linear operators, and so try to answer the question: what are the asymptotics of the best Fourier methods, linear methods, methods of approximation by subspaces, and non-linear methods, in relation to the smoothness of the class being approximated?

7.2. What is smoothness?

There are widely differing approaches to the definition of this concept. In previous sections we have come across the Sobolev and Nikol'skii classes. There are also the Besov classes and many others. It is impossible to discuss all these questions here. We shall restrict ourselves to the simplest situations. In the one-dimensional case, the simplest version of smoothness is characterized by a pair $(1/p, \alpha)$ (where p and α are numbers characterizing the "metric" and the "number of derivatives", as was the case for, say, the Sobolev and Nikol'skii classes $W_p^\alpha(\mathbb{T})$ and $H_p^\alpha(\mathbb{T})$) and thus is determined by a point in the strip $I \times \mathbb{R}$.

In the multidimensional case the definition of smoothness involves the various partial derivatives. For example, the definition of the Nikol'skii classes $H_p^\alpha(I^n)$ involves all the derivatives $\partial^r / \partial t_1^{r_1} \dots \partial t_n^{r_n}$, where $\sum r_i = r$, that is, the "set" of derivatives. We may also consider cases when each derivative is given in its own metric. Thus here it is natural to regard smoothness as a subset $\{(1/p^i, \alpha^i)\}$ of $I \times \mathbb{R}^n$, $1 \leq p^i \leq \infty$, $\alpha^i \in \mathbb{R}^n$.

We shall also apply this point of view in the one-dimensional case, that is, we shall characterize smoothness by a subset (in general infinite) of $I \times \mathbf{R}$.

Let us consider the one-dimensional case in more detail. Here the situation has now been considerably clarified. The general formulation is as follows. Suppose we know that the functions to be approximated are characterized by a finite subset $\{1/p^i, \alpha^i\}_{i=1}^n$ of $I \times \mathbf{R}$. This means that we shall consider the intersection of the Nikol'skii classes $H_p^{\alpha^i}(\mathbf{T})$. The approximations will be considered in the metric $L_p(\mathbf{T})$. The questions is: what are the asymptotics of the Fourier diameter, or of the linear, Kolmogorov, and Aleksandrov diameters of this set of functions, and also what is its ε -entropy? This is one of the central questions on "smoothness and approximation". We shall prove a relevant theorem in this article, and formulate some assertions in the multidimensional case, where the problems have not all been completely solved yet.

Let us say a little about the development of this topic. As we said at the beginning, the greater part of the work in approximation theory has in fact been concerned with it. There are, of course, the fundamental results of Bernstein, Jackson, Hardy-Littlewood, Riesz, Fejér, de la Vallée Poussin, Nikol'skii, Dzyadyk, Korneichuk, and others, concerned with the "second" stage of approximation theory, when approximation was understood to mean approximation of classes of functions by polynomials and rational functions. To this we must also undoubtedly refer the trend concerned with embedding theorems. A conspicuous role here has been played by the ideas of Nikol'skii, who united the theories of embedding and approximation. The whole of the "third" period, when diameters began to be studied, is also concerned with this theme.

As we have said, the initial results here were obtained by Kolmogorov in 1936, then by Stechkin in 1954, considering the pairs $(0, \alpha)$ in $C(\mathbf{T})$ and $(1/2, \alpha)$ in $L_2(\mathbf{T})$. Then Kolmogorov's work (1955–1959) on ε -entropy appeared, where the outlines of the problem were fully defined. The work of Babenko was of great importance in the development of the whole topic. His first results were published in 1960. He always emphasized Kolmogorov's role in forming the attitudes which in Babenko's opinion must predominate in working out the theoretical foundations of computational mathematics. From 1960 very many papers appeared. In the Steklov Institute the problem was actively studied by Kashin, Oskol'kov, Telyakovskii, and Temlyakov, and at Moscow State University it was the subject of my research seminar over many years; those taking part included Buslaev, Galeev, Din Zung, Magaril-Il'yaev, Maiorov, Nguyễn Thị Thiệu Hoa, Stesin, and Khodulev. The researches of Mityagin, Osipenko, Smolyak, Farkov, and others must also be mentioned. These questions have been intensively studied in Leningrad, Kiev, Dnepropetrovsk, Yaroslavl', and Alma-Ata, and there are many foreign investigators, most of whom are named elsewhere in

this article. There are many monographs on the subject: [17], [27], [34], [35].

We shall begin our survey with the simplest version in which smoothness is specified by a single point $(1/p, r)$ in the strip $I \times \mathbf{R}$. More precisely, we consider approximations to the classes $H_p^r(\mathbf{T})$ in the metric $L_q(\mathbf{T})$.

First let us turn briefly to the classical theorems given above. The first of them (the Hardy-Littlewood theorem) studies, in particular, the problem of embedding $H_p^r(\mathbf{T})$ in $L_q(\mathbf{T})$ for distinct p and q , and in the remaining cases studies approximations of the pairs $(H_p^r(\mathbf{T}), L_p(\mathbf{T}))$ in the case $p = q$. Therefore we first describe the general situation by answering the question: how do the spaces $\mathcal{T}_n$ approximate (by the Fourier or Fejér methods) to the class $H_p^r(\mathbf{T})$ in the space $L_q(\mathbf{T})$ for arbitrary p and q ?

Theorem 1. *Let $1 \leq p, q \leq \infty$. Then the following assertions are true:*

a) (on embedding) *the class $H_p^r(\mathbf{T})$ can be compactly embedded in $L_q(\mathbf{T})$ if and only if $\varphi(r, 1/p, 1/q) = r - (1/p - 1/q)_+ > 0$;*

b) (approximation by $\mathcal{T}_n$) *if $\varphi(r, 1/p, 1/q) < 0$, then*

$$d(H_p^r(\mathbf{T}), \mathcal{T}_n, L_q(\mathbf{T})) \asymp k^{-\varphi(r, 1/p, 1/q)};$$

c) (approximation by Fourier sums)

$$\sup_{x(\cdot) \in H_p^r(\mathbf{T})} \|x(\cdot) - S_k x(\cdot)\|_{L_q(\mathbf{T})} \asymp \begin{cases} k^{-\varphi(r, 1/p, 1/q)}, & (1/p, 1/q) \neq (0, 0) \vee (1, 1), \\ k^{-r} \log k, & (1/p, 1/q) = (0, 0) \vee (1, 1); \end{cases}$$

d) (approximation by Fejér sums), $1 < p, q < \infty$;

$$\sup_{x(\cdot) \in H_p^r(\mathbf{T})} \|x(\cdot) - \mathcal{F}_k x(\cdot)\|_{L_q(\mathbf{T})} \asymp \begin{cases} k^{-\varphi(r, 1/p, 1/q)}, & \varphi(r, 1/p, 1/q) < 1, \\ k^{-1}, & \varphi(r, 1/p, 1/q) > 1. \end{cases}$$

(The case $\varphi(r, 1/p, 1/q) = 1$ has also been analysed, but we shall not go into this here.)

Assertion a) supplements the Hardy-Littlewood theorem, b) the Jackson-Bernstein theorem, c) the Riesz-Lebesgue theorem, and d) the Fejér-Bernstein theorem.

Note that the same function φ characterizing the relation between smoothness and approximation appears in all these theorems.

We now consider the Kolmogorov approach to the same topic and try to explain what are the best methods (linear and non-linear) of approximation of the same pair $(H_p^r(\mathbf{T}), L_q(\mathbf{T}))$.

In order to state the result we must introduce the following notations:

$$\begin{aligned} \lambda(r, 1/p, 1/q) &= \{r - (1/p - 1/q)_+, 1/p \leq 1/q \vee 1/2 \leq 1/p, 1/q \leq 1/p; \\ &\quad r - (1/p - 1/2), 1/p' \leq 1/q \leq 1/2, 1/q \leq 1/p; r - (1/2 - 1/q), \\ &\quad 1/2 \leq 1/p', 0 \leq 1/q \leq 1/p'; r - (1/p - 1/q), \\ &\quad 1/q \leq 1/2, 1/q \leq 1/p\}; \end{aligned}$$

and

$$\begin{aligned} d(r, 1/p, 1/q) &= \{r, 1/p \leq 1/q; r - (1/p - 1/q), 1/2 \leq 1/p, 1/q \leq 1/p; \\ &\quad r - (1/p - 1/2), 1/2 \leq 1/p, 1/q \leq 1/2\}, \\ a(r, 1/p, 1/q) &= r. \end{aligned}$$

Theorem 2. If $\varphi(r, 1/p, 1/q) > 0$, then

- (1) $\varphi_k(H_p^r(\mathbf{T}), L_q(\mathbf{T})) \asymp k^{-\varphi(r, 1/p, 1/q)} \quad (1 \leq p, q < \infty),$
- (2) $\lambda_k(H_p^r(\mathbf{T}), L_q(\mathbf{T})) \asymp k^{-\lambda(r, 1/p, 1/q)} \quad (1 \leq p, q \leq \infty),$
- (3) $d_k(H_p^r(\mathbf{T}), L_q(\mathbf{T})) \asymp k^{-d(r, 1/p, 1/q)} \quad (1 \leq p, q \leq \infty),$
- (4) $a_k(H_p^r(\mathbf{T}), L_q(\mathbf{T})) \asymp k^{-a(r, 1/p, 1/q)} \quad (1 \leq p, q \leq \infty),$
- (5) $\mathcal{E}_k(H_p^r(\mathbf{T}), L_q(\mathbf{T})) \asymp e^{-1/r}.$

Exactly the same formulae hold for the Sobolev classes $W_p^r(\mathbf{T})$ and the non-periodic Nikol'skii classes $H_p^r(I)$.

Note that besides the function φ , formulae (1)–(4) involve three more functions, that is, *each method of approximation gives rise to its own kind of asymptotics*. We can now compare them.

Comparing (1) with assertion b) of Theorem 1, we see that the space $\mathcal{T}_n$ is optimal (for $1 < p, q < \infty$) for the Fourier method, while the Fejér method is optimal only for small smoothnesses (see d)), when φ is less than 1. Later “saturation” occurs, and the method no longer reacts to a further increase in smoothness. Comparison of c) with (3) shows that the trigonometric polynomials (with $(1/p, 1/q) \neq (0, 0) \vee (1, 1)$ this is simply the Fourier method) is asymptotically optimal in the sense of the Kolmogorov diameter. But in the trapezium III $\cup$ IV $\cup$ V, the classical methods successively give way to the optimal methods. It is still not entirely clear how to describe the optimal methods explicitly for points $(1/p, 1/q)$ in this trapezium. At present we can only say in the most general terms that in III the best subspaces and the best linear method give asymptotically identical results, but in IV the linear method gives way to the best (non-linear) method of approximation by subspaces. In II–IV approximations by subspaces give way to approximations by non-linear objects of the same dimension, and in V for some reason they give asymptotically the same approximations.

Formulae (1)–(5) were obtained as the result of lengthy efforts; definitive results for formulae (1)–(3) were obtained respectively by Temlyakov, Maiorov, and Kashin. The first real progress was made by Ismagilov, the most important result (3) was obtained by Kashin, overcoming great difficulties, and (5) was obtained by Birman and Solomyak.

In several dimensions the results are very similar to those in the one-dimensional case. Here the “single point” is in the space $I \times \mathbf{R}^n$. The similarity between the one-dimensional and multidimensional cases is most

complete for the Sobolev classes. Let $r = (r_1, \dots, r_n)$, $0 < r_1 = \dots = r_{l+1} < r_{l+2} \leq \dots \leq r_n$. Then the following formulae hold for the pair $(\dot{W}_p^r(\mathbf{T}^n), L_q(\mathbf{T}^n))$:

$$\varphi_h \asymp (k^{-1} \log^l k)^{\Phi(r_1, 1/p, 1/q)}, \quad \lambda \asymp (k^{-1} \log^l k)^{\lambda(r_1, 1/p, 1/q)}, \\ d_h \asymp (k^{-1} \log^l k)^{d(r_1, 1/p, 1/q)}.$$

Here we consider the pair $(\dot{W}_p^r(\mathbf{T}^n), L_q(\mathbf{T}^n))$, $1 < p, q < \infty$, where $\dot{W}_p^r(\mathbf{T})$ is the set of functions $x(\cdot)$ for which the Fourier coefficient $x_k = 0$

whenever $k = (k_1, \dots, k_n)$ and $\prod_{i=1}^n k_i = 0$, and also the Weyl derivative of order $r = (r_1, \dots, r_n) \in \mathbf{R}_+^n$ has norm bounded by 1 in $L_p(\mathbf{T}^n)$.

Babenko's papers at the beginning of the 1960's initiated research into this kind of problem. It was continued by Mityagin and Telyakovskii. The results stated above were largely obtained by the efforts of Galeev ($q \leq p$) and Temlyakov ($q \geq p$). The problem has been solved completely for the Fourier diameters of the Nikol'skii classes, but there still remain blank spots for the other diameters. There are still more questions about the fundamental problem when the intersection of classes is considered. We shall give the definitive result in the one-dimensional case. Suppose the smoothness is given by the set $\Gamma = \{(1/p^i, r^i)\}_{i=1}^m$, $W^\Gamma(\mathbf{T}) = \bigcap_{i=1}^m W_p^{r^i}(\mathbf{T})$, $1 < p^i < \infty$, $r^i > 0$. Consider the set

$$\mathfrak{G}_i = \{(1/q, \rho) \in I \times \mathbf{R} \mid \rho \leq \Phi(r^i, 1/p^i, 1/q)\}, \\ \mathfrak{G}(\Gamma) = \text{co} \bigcup_{i=1}^m \mathfrak{G}_i, \quad \Phi(\Gamma, 1/q) = \sup \{\rho \mid (1/q, \rho) \in \mathfrak{G}(\Gamma)\}, \\ d(\Gamma, 1/q) = \begin{cases} \Phi(\Gamma, 1/q), & 1/2 \leq 1/q \leq 1, \\ \Phi(\Gamma, 1/2), & 0 \leq 1/q \leq 1/2, \end{cases} \quad a(\Gamma, 1/q) = \max \{r^1, \dots, r^m\}.$$

Then we have the following formulae, obtained by Galeev:

$$\varphi_h(W^\Gamma(\mathbf{T}), L_q(\mathbf{T})) \asymp k^{-\Phi(\Gamma, 1/q)}, \\ d_h(W^\Gamma(\mathbf{T}), L_q(\mathbf{T})) \asymp k^{-d(\Gamma, 1/q)}, \\ a_h(W^\Gamma(\mathbf{T}), L_q(\mathbf{T})) \asymp k^{-a(\Gamma, 1/q)}.$$

The n -dimensional case will apparently involve certain functionals on the polytopes $\text{co}\{(1/p^i, r^i)\}$ that specify the smoothness. At present their form is still not clear.

These are the kind of results that have been obtained from the formulation, going back to Kolmogorov, of the problems on the relations between smoothness and approximation applied to differentiable functions.

Kolmogorov's theorem on the entropy of analytic functions has had a rather different fate. Kolmogorov repeatedly emphasized that finding the

exact asymptotics of the ϵ -entropy would enable new, interesting and important techniques in analytic function theory to be discovered. The story later developed as follows. First Vitushkin obtained partial results showing that in the "consistent cases" such as, for example, approximation of functions in $\mathcal{H}_\infty(I)$ in the metric $C(D_R)$, for $R < 1$ segments of Taylor series turned out to be the optimal method (it was necessary to "store" the coefficients of the series with the necessary accuracy in order to obtain the required ϵ -net). Then the general case (considering the pair $(A^G, C(T))$, where G is a simply-connected domain, and T a continuum contained in G and not separating G —for the notations see §1.6) was simultaneously taken up by Babenko and Erokhin who simultaneously obtained the same result by entirely different methods: Erokhin found as it were an analogue of the Taylor series for the pair (G, T) , while Babenko constructed approximations by rational functions. After a while it became clear that the old methods of rational approximation, already worked out by Walsh in the 1930's, gave the necessary asymptotics for multiply-connected domains and multiply-connected continua (Levin-Tikhomirov, Widom, and others). Then Farkov proved that the special bases, also constructed at the beginning of the century, have in this general "multiply-connected" problem the solution which Erokhin obtained in the "simply-connected" case.

Thus the interesting new methods of approximation turned out in this case to be in fact the "old" ones.

Multidimensional problems here still await solution.

Let me add a few more words to what has been said. Kolmogorov was responsible for fundamental mathematical results in approximation theory which have provided motivation for the whole area. We have tried to show how the various topics have developed since. But in Kolmogorov's own work, the same results led him to completely different goals. Thus, for example, the theorem on ϵ -entropy of smooth functions led him to the explanation of the root causes underlying Vitushkin's theorem on the non-representability of smooth functions of several variables by the superposition of smooth functions of fewer variables (with certain relations between the smoothnesses and the number of variables). This in turn led him to consideration of Hilbert's 13th problem and to its complete solution. Kolmogorov's result on ϵ -entropy of analytic functions suggested the idea of "approximate dimension" to him, which led to the proof that spaces of analytic functions of differing numbers of variables were not linearly isomorphic.

A similar breadth of outlook has always characterized Kolmogorov's work.

In conclusion we must again recall his paper [3] No. 8, p.199. In it Kolmogorov proposed to rise one step higher and along with the *precision* of an approximation to consider the *complexity* of achieving it. For simplicity he restricted himself to continuous functions $x(\cdot)$ not exceeding 1 ($x(\cdot) \in BC(I)$). Let s be sufficiently large. Then each table that associates

with the binary number $t = 0 \cdot t_1 \dots t_s$ ($t_i \in \{0, 1\}$) a binary number $x_\epsilon = -1 + x_0 \cdot x_1 \dots x_s$ ($x_i \in \{0, 1\}$) satisfying the condition $|x(t) - x_\epsilon(t)| \leq \epsilon/2$ can be regarded as a table giving the values of $x = x(t)$ with precision ϵ for any $t \in I$.

The function $t = 0 \cdot t_1 \dots t_s \rightarrow x_\epsilon = -1 + x_0 \cdot x_1 \dots x_s$ can be interpreted as a discrete function $X = \Phi(T)$, mapping vectors $T = (t_1, \dots, t_s)$ to vectors $X = (x_0, \dots, x_s)$, where the variables t_i and x_j take only the values $\{0, 1\}$. Discrete vector functions—"functions of the algebra of logic"—can be characterized by *complexity* (definable in various ways), which, for example, may signify the minimum number of terms in the representation of $\Phi(\cdot)$ as the superposition of 16 elementary functions.

The minimum complexity of functions Φ corresponding to tables giving $x(\cdot)$ with precision ϵ is naturally called the complexity of the ϵ -definition of $x(\cdot)$. The complexity of a class is denoted by $K_\epsilon(W)$.

The following result is valid.

Theorem (Yu. Ofman). $K_\epsilon(W_\infty^r(I), C(I)) \asymp (1/\epsilon)^{1/r} \log 1/\epsilon$.

At present there are very few additional theorems of this kind.

§8. Proofs

In this section we shall give proofs of some of the theorems stated earlier so that the reader can become acquainted with the *methods* developed in this part of approximation theory. The proofs will be carried out in a *formalized* way, where each implication (equality, inequality, inclusion, and so on) is justified either by reference to an already established formula, or by an indication that it holds by definition (written as def), or after the identity mapping Id, or by a well-known or easily verifiable property listed in para. 0 of the proof, where all the basic facts used later are collected. The notations of the sections in which the theorem being proved was stated are freely used. Since we want to demonstrate the basic ideas of the proof, we sometimes restrict ourselves to the simplest special cases.

8.1. Proof of Theorem 2 of §2 (Nikol'skii) for $r = 0$. Let

$$\begin{aligned} h &:= \pi/(2n+1); \quad z_k := kh; \quad 1 \leq k \leq 2n+1; \\ \alpha_k &:= \int_{z_k}^{z_{k+1}} \mathcal{D}_n(t) dt, \quad 1 \leq k \leq 2n; \quad \Delta_s := \int_{z_{2s-1}}^{z_{2s}} \mathcal{D}_n(t) dt, \quad 1 \leq s \leq n; \\ \gamma_s &:= \sum_{m=s}^n \Delta_m; \quad \xi(t) := x(t) - x(z_{2s}), \quad z_{2s-1} < t \leq z_{2s}; \\ \mu(t) &= (2 \sin sh)^{-1}, \quad z_{2s-1} < t \leq z_{2s+1}; \\ \mathcal{D}_n^*(t) &:= \mu(t) \sin(n+1/2)t, \quad \beta_{n\omega} := \int_0^{\pi/2} \omega(4\tau/(2n+1)) \sin \tau d\tau. \end{aligned}$$

Note that $\mathcal{D}_n(t) = 2^{-1} \sin(n + 1/2)t / \sin t/2$ changes sign at the points z_{2s} .

$$0.1. \quad x(\cdot) \in H^\omega(\mathbf{T}), \quad x(0) = 0 \Rightarrow \int_0^h \mathcal{D}_n(t) x(t) dt = O(\omega(h)).$$

$$0.2. \quad \alpha_k > \alpha_{k+1}, \quad 1 \leq k \leq 2n; \quad |\Delta_s| > |\Delta_{s+1}|, \quad 1 \leq s \leq n-1, \\ \text{sign } \Delta_s = (-1)^s, \quad |\gamma_s| < |\Delta_s| = \alpha_{2s-1} - \alpha_s.$$

$$0.3. \quad \int_h^\pi |\mathcal{D}_n(t) - \mathcal{D}_n^*(t)| = O(1).$$

$$0.4. \quad \sum_{s=1}^n (2 \sin sh)^{-1} = \pi n \log n + O(n).$$

$$0.5. \quad \varphi_{n0\omega} := \sup \left\{ 2\pi^{-1} \int_0^\pi \mathcal{D}_n(t) x(t) dt \mid x(\cdot) \in H^\omega(\mathbf{T}), \quad x(0) = 0 \right\}.$$

All these facts are easy to prove directly. For more details see [12].

1. Upper bound.

Let $x(\cdot) \in H^\omega(\mathbf{T})$, $x(0) = 0$. Then

$$(1) \quad \int_0^\pi \mathcal{D}_n(t) x(t) dt \stackrel{0.1}{=} \int_h^\pi \mathcal{D}_n(t) x(t) dt + O(\omega(h)) \stackrel{\text{Id}}{=} \\ \stackrel{\text{Id}}{=} \int_h^\pi \mathcal{D}_n(t) \xi(t) dt + \sum_{s=1}^n x(z_{2s}) \Delta_s + O(\omega(h)) =: A(x(\cdot)) + B(x(\cdot)) + O(\omega(h)).$$

Applying the Abel transformation (Ab) to $B(x(\cdot))$, we obtain

$$|B(x(\cdot))| \stackrel{\text{Ab}}{\leq} |\gamma_1 x(z_2)| + \sum_{s=2}^n |\gamma_s| |x(z_{2s-2}) - x(z_{2s})| \leq \\ \leq \omega(2h) \sum_{s=1}^n |\gamma_s| \stackrel{0.2}{\leq} \omega(2h) \alpha_1 = O(\omega(h)),$$

$$|Ax(\cdot)| \stackrel{\text{def}}{=} \left| \int_h^\pi \mathcal{D}_n(t) \xi(t) dt \right| \stackrel{0.3}{=} \left| \int_h^\pi \mathcal{D}_n^*(t) \xi(t) dt \right| + O(\omega(h)) \stackrel{\text{Id}}{=} \\ \stackrel{\text{Id}}{=} \left| \sum_{s=1}^n (2 \sin sh)^{-1} \int_{z_{2s-1}}^{z_{2s+1}} \xi(t) \sin(n + 1/2)t dt \right| + O(\omega(h)) = \\ = \left| \sum_{s=1}^n (2 \sin sh)^{-1} \int_{z_{2s-1}}^{z_{2s}} (\xi(t) - \xi(2z_{2s} - t)) \sin(n + 1/2)t dt \right| + O(\omega(h)) \leq \\ \leq \left| \sum_{s=1}^n (2 \sin sh)^{-1} n^{-1} \beta_{n\omega} \right| + O(\omega(h)) = \pi^{-1} \log n \beta_{n\omega} + O(\omega(h)).$$

2. For the lower bound we put

$$x_{n\omega}(t) = \{0, t \in [0, z_2]; (-1)^s 2^{-1\omega} (2(t - z_{2s})), t \in [z_{2s}, z_{2s+1}], 1 \leq s \leq n; \\ (-1)^s 2^{-1\omega} (2(z_{2s-2} - t)), t \in [z_{2s+1}, z_{2s+2}], 1 \leq s \leq n\},$$

and show, as in para. 1, that

$$A(x_{n\omega}(\cdot)) \geq \pi^{-1} \log n \beta_{n\omega} + O(\omega(h)), \quad B(x_{n\omega}(\cdot)) = 0, \\ \Rightarrow \varphi_{n0\omega} \geq 2\pi^{-2} \log n \beta_{n\omega} + O(\omega(h)), \quad (1) \Rightarrow \varphi_{n0\omega} = 2\pi^{-2} \log n \beta_{n\omega} + O(\omega(h)). \quad \triangleright$$

8.2. Proof of Theorem 3 of §2 (Dzyadyk).

$$0.1. \quad x(\cdot) \in W_\infty^r(\mathbb{T}) \Rightarrow x(t) = c - \pi^{-1} \int_{\mathbb{T}} \varphi_r(t - \tau) h(\tau) d\tau,$$

$$h(\tau) = x^{(r)}(\tau + \pi), \quad \varphi_r(t) = \sum_{k \in \mathbb{N}} (-1)^k k^{-r} \cos(kt - \pi/2);$$

$$0 < r < 1 \Rightarrow \dot{\varphi}_r(t) = -\pi(1-r)(\Gamma(r))^{-1} \sum_{k \in \mathbb{N}} (\pi(2k-1) + t)^{r-2} [8,36].$$

0.2. The system of functions $\{y_k(\cdot)\}_{k=1}^n$, $y_k(\cdot) \in C^n([t_0, t_1])$ is called an *ET-system (generalized Chebyshev system)* if any linear combination of them has no more than $n-1$ zeros (counted according to multiplicity) on $[t_0, t_1]$, and an *EM-system (in honour of Markov)* if for any s , $1 \leq s \leq n$, the system $\{y_k(\cdot)\}_{k=1}^s$ is an *ET-system*. Let $\{y_k(\cdot)\}_{k=1}^n$ be an *ET-system* on (t_0, t_1) , $[t'_0, t'_1] \subset (t_0, t_1)$, $\tau \in (\tau_0, \tau_1) \setminus [t'_0, t'_1]$. Let $z_s(\cdot) \in \text{lin} \{y_k(\cdot)\}_{k=1}^n$, $z_s^{(i)}(\tau) = 0$, $0 \leq i \leq n-s$, $1 \leq s \leq n-1$, $z_s^{(n-s+1)}(\tau) \neq 0$, $z_n(\tau) \neq 0$. Then, clearly, the system $\{z_s(\cdot)\}_{s=1}^n$ is an *EM-system* on (t'_0, t'_1) .

0.3. Suppose we are given a differential operator

$$Lx(t) = x^{(m)}(t) + \sum_{h=1}^m a_h(t) x^{(m-h)}(t),$$

and $\{z_s(\cdot)\}_{s=1}^m$ belonging to the kernel of L and forming an *EM-system* on (t_0, t_1) . Then the following equality holds on this interval:

$$Lx(t) = w_{m+1}(t) D(w_m(t) D(\dots (Dw_1(t) x(t)) \dots)), \quad D := \frac{d}{dt}, \\ w_i(\cdot) \in C^{n-i+1}((t_0, t_1)), \quad w_i(\cdot) > 0 [37].$$

$$0.4. \quad \int_{\mathbb{T}} y(t) \text{sign} \sin nt \, dt = 0 \text{ for all } y(\cdot) \in \mathcal{T}_{n-1},$$

$$\pi^{-1} \|\varphi_r * \text{sign} \sin n \cdot\|_{L_\infty(\mathbb{T})} = n^{-r} K_r.$$

0.5. Let G be a function continuous on $(-\pi, \pi)$ and summable on $\mathbb{T}$, $\lambda \in \mathbb{T}$, and let $T_{n-1}(\cdot) \in \mathcal{T}_{n-1}$ be a polynomial such that $G(\cdot) - T_{n-1}(\cdot)$ vanishes at the points $\lambda + k\pi/n$, $0 \leq k \leq 2n-2$. Then a necessary and

sufficient condition for $G(\lambda + (2n-1)\pi/n) = T(\lambda + (2n-1)\pi/n)$ is that

$\sum_{k=0}^{2n} (-1)^k G(\lambda + k\pi/n) = 0$ [38]. Since the function on the left-hand side has mean zero, there must be a point where it changes sign. Thus there is a number θ and a polynomial $T_{nr}(\cdot) \in \mathcal{T}_{n-1}$ such that $T_{nr}(\cdot)$ interpolates $\varphi_r(\cdot)$ at the points $\theta + k\pi/n$, $0 \leq k \leq 2n-1$.

1. Let $Z(A, x(\cdot))$ denote the number of zeros of $x(\cdot)$ in the set A counted according to multiplicity.

Lemma (Dzyadyk). a) $Z((-\pi, \pi), \varphi_r(\cdot) - T(\cdot)) \leq 2n-1$ for all $T(\cdot) \in \mathcal{T}_{n-1}$,

b) $Z(\mathbf{T}, \varphi_r(\cdot) - T(\cdot)) \leq 2n$ for all $T(\cdot) \in \mathcal{T}_{n-1}$, $r > 1$.

We give a proof following Nguyễn Thị Thiệu Hoa.

a) The functions $\{\cos k\cdot, \sin k\cdot\}_{k=0}^{n-1}$ belong to the kernel of the operator $L_{n-1} := D \prod_{k=1}^{n-1} (D^2 + k^2)$ and form an ET -system on $(-\pi, \pi)$, so that for L_{n-1} the representation 0.3 holds. Applying L_{n-1} to the function $\varphi_r(\cdot) - T(\cdot)$, and using the fact that $L_{n-1}T(\cdot) = 0$, and $\varphi_r(\cdot) < 0$, $\varphi_r^{(2i-1)}(\cdot) \leq 0$, $i \in \mathbf{N}$, which follows from 0.1, we obtain

$$0 = Z((-\pi, \pi), L_{n-1}\varphi_r(\cdot)) = Z((-\pi, \pi), L_{n-1}(\varphi_r(\cdot) - T(\cdot))) \geq \\ \geq Z((-\pi, \pi), \varphi_r(\cdot) - T(\cdot)) - 2n + 1.$$

b) Let $r = r_1 + r_2$, $r_1 \in \mathbf{N}$, $0 < r_2 \leq 1$. We have (from Rolle's theorem again)

$$Z(\mathbf{T}, \varphi_{r_1+r_2}(\cdot) - T(\cdot)) \leq Z(\mathbf{T}, \varphi_{r_2}(\cdot) - T^{(r_1)}(\cdot)) \leq \\ \leq Z((-\pi, \pi), \varphi_{r_2}(\cdot) - T^{(r_1)}(\cdot)) + 1 \stackrel{(a)}{\leq} 2n.$$

From the lemma and 0.5 it follows that $\varphi_r(\cdot) - T_{nr}(\cdot)$ changes sign at the points $\theta + k\pi/n$, $0 \leq k \leq 2n-1$, and only at these points; also, for $0 < r \leq 1$, $\theta = \pi$.

2. (Favard's basic argument.) Let $x(\cdot) \in W_\infty^r(\mathbf{T})$,

$$y(t) = \pi^{-1} \int_{\mathbf{T}} T_{nr}(t-\tau) x^{(r)}(\tau+\pi) d\tau \Rightarrow y(\cdot) \in \mathcal{T}_{n-1}, \quad |x(t) - y(t)| \stackrel{0.1}{=} \\ = \pi^{-1} \left| \int_{\mathbf{T}} (\varphi_r(t-\tau) - T_{nr}(t-\tau)) x^{(r)}(\tau+\pi) d\tau \right| \stackrel{1.}{=} \\ = \pi^{-1} \left| \int_{\mathbf{T}} (\varphi_r(t) - T_{nr}(t)) \operatorname{sign} \sin n(t-\theta) dt \right| \stackrel{0.4}{=} n^{-r} K_r.$$

3. For the lower bound we must note that the function $\tilde{x}_{nr}(t) := \pi^{-1} \times \int_{\mathbf{T}} \varphi_r(t-\tau) \operatorname{sign} \sin n\tau d\tau$ has $2n$ -alternance, and thus

$$d(\tilde{x}_{nr}(\cdot), \mathcal{T}_{n-1}, \mathcal{C}(\mathbf{T})) = n^{-r} K_r \triangleright.$$

8.3. **Theorem 4** of §2 (Korneichuk) for $r = 0$.

$$0.1. \quad x(\cdot) \in W_{\infty}^1(\mathbf{T}) \Rightarrow \exists T_n(\cdot) \in \mathcal{T}_{n-1} : \|x(\cdot) - T_n(\cdot)\|_{C(\mathbf{T})} \leq \pi/2n$$

(Favard's theorem for $r = 1$).

0.2. Each concave function has a support line at any point.

0.3. Chebyshev's theorem on alternance.

◁ 1. **Lemma** (Korneichuk). Let $x(\cdot) \in H^{\omega}(\mathbf{T})$, $K > 0$; then there is a function $\hat{y}(\cdot) \in KW_{\infty}^1(\mathbf{T})$ ($\Leftrightarrow \hat{y}(\cdot) \in AC(\mathbf{T})$, $|\dot{\hat{y}}(t)| \leq K$ almost everywhere) such that

$$(1) \quad \|x(\cdot) - \hat{y}(\cdot)\|_{C(\mathbf{T})} \leq \alpha_{\omega K} := 2^{-1} \max_{t>0} (\omega(t) - Kt).$$

Let

$$y(t) = \sup \{z(t) \mid z(\cdot) \in KW_{\infty}^1(\mathbf{T}), z(t) \leq x(t) \text{ for all } t\}.$$

It is easy to see that $y(\cdot) \in KW_{\infty}^1(\mathbf{T})$ (and hence $\dot{y}(\cdot)$ exists a.e.). If we assume that the inequality $|\dot{y}(\tau)| < K$ is satisfied at a point τ , where $y(\tau) < x(\tau)$, then (assume $0 \leq \dot{y}(\tau) < K$) the function $\tilde{y}_{\varepsilon}(t) := \max(y(t), y(\tau) + \varepsilon - K|t - \tau|)$ for small ε will have a greater value at τ than $y(\cdot)$, which is impossible, that is, the set $\{t \mid y(t) < x(t)\}$ is a system of intervals $\{(\tau_k, \tau_{k+1})\}$, in which $\dot{y}(t) \equiv K$ (or $\dot{y}(t) \equiv -K$), that is,

$$|x(t) - y(t)| = |x(t) - x(\tau_k) - K(t - \tau_k)| \stackrel{\text{def}}{\leq} \omega(t - \tau) - K(t - \tau) \stackrel{\text{def}}{\leq} 2\alpha_{\omega K}.$$

It only remains to put

$$\hat{y}(t) = y(t) + 2^{-1} \|x(\cdot) - y(\cdot)\|_{C(\mathbf{T})}. \triangleright$$

2. Let us take the support line $\omega(\pi/n) + K(t - \pi/n)$ to $\omega(\cdot)$ at the point π/n :

$$(2) \quad \Leftrightarrow 2^{-1} \max_{t>0} (\omega(t) - Kt) = 2^{-1} (\omega(\pi/n) - K\pi/n) \quad (\text{see } 0.2).$$

Let $\hat{y}(\cdot)$ be the function in the lemma (for $x(\cdot) \in H^{\omega}(\mathbf{T})$), and $\hat{T}_n(\cdot)$ the Favard polynomial for $\hat{y}(\cdot)$ (see 0.1). Then

$$\begin{aligned} |x(t) - \hat{T}_n(t)| &\leq |x(t) - \hat{y}(t)| + |\hat{y}(t) - \hat{T}_n(t)| \stackrel{(1), (2), 0.1}{\leq} \\ &\stackrel{(1), (2), 0.1}{\leq} 2^{-1} (\omega(\pi/n) - K\pi/n) + K\pi/2n = 2^{-1} \omega(\pi/n). \end{aligned}$$

For the lower bound it is enough to note that the function $\tilde{x}_{n0\omega}(\cdot)$, constructed in §2.2, has $2n$ -alternance, and hence

$$d(\tilde{x}_{n0\omega}(\cdot), \mathcal{T}_{n-1}, C(\mathbf{T})) \stackrel{0.3}{=} \|\tilde{x}_{n0\omega}(\cdot)\|_{C(\mathbf{T})} = 2^{-1} \omega(\pi/n). \triangleright$$

8.4. **Theorem 6** of §2 (Babenko-Taikov).

◁ Let $\alpha_k^r = (k(k-1) \dots (k-r+1))^{-1}$.

0.1. If $x(\cdot) \in \mathcal{B}(\mathcal{D}) \cap L_p(\partial\mathcal{D})$, $x(z) = \sum_{h \in \mathbf{Z}_+} x_h z^h$, $\eta(z) = \sum_{h=0}^{n-1} \eta_h z^h$, then

there is a $y(\cdot) \in \mathcal{P}_{n-1}$ such that

$$\begin{aligned} (1) \quad & x(e^{it}) - y(e^{it}) = \\ & = \pi^{-1} \int_{\mathbf{T}} (x^{(r)}(e^{i(t+\tau)}) - \eta(e^{i(t+\tau)})) e^{-i(n-r)\theta} \left(\left(\frac{\alpha_n^r}{2} + \sum_{h=1}^{\infty} \alpha_{n+h}^r \cos k\tau \right) \right) d\tau. \end{aligned}$$

This identity can be verified by direct integration.

0.2. $(\alpha_n^r/2) + \sum_{h \in \mathbf{N}} \alpha_{n+h}^r \cos k\tau \geq 0$ for all $\tau \in \mathbf{T}$ (proved by a double

application of the Abel transformation).

0.3. Minkowski's inequality.

1. Upper bound. Let $\eta(e^{it})$ be the element of best approximation to $x^{(r)}(e^{it})$ in the space $\text{lin} \{e^{ik}; 0 \leq k \leq n-1\}$ in $L_p(\mathbf{T})$. Then

$$\begin{aligned} \|x(\cdot) - y(\cdot)\|_{L_p(\partial\mathcal{D})} &\stackrel{(1), 0.2, 0.3}{\leq} \\ &\stackrel{(1), 0.2, 0.3}{\leq} \|x^{(r)}(\cdot) - \eta(\cdot)\|_{L_p(\partial\mathcal{D})} \pi^{-1} \int_{\mathbf{T}} \left(\frac{\alpha_n^r}{2} + \sum_{h \in \mathbf{N}} \alpha_{n+h}^r \cos k\tau \right) d\tau \leq \alpha_n^r. \end{aligned}$$

2. The lower bound is attained by the function z^n .

8.5. **Theorem 1** of §3 (Tikhomirov) and **Theorem 5 a)** of §3 (Babenko-Tikhomirov-Taikov).

0.1. Favard's theorem (see (1) in §2). 0.2. Theorem 6 of §2 (Babenko-Taikov). 0.3. Borsuk's theorem: $f \in C(\mathbf{S}^n, \mathbf{R}^n)$, $f(x) = -f(-x) \Rightarrow$ there is an $\hat{x}$ such that $f(\hat{x}) = 0$. 0.4. Rolle's theorem. 0.5. The Bernstein-Zygmund inequality:

$$x(t) = \sum_{|k| \leq n} x_k \times \exp(ikt) \Rightarrow \|\dot{x}(\cdot)\|_{L_p(\mathbf{T})} \leq n \|x(\cdot)\|_{L_p(\mathbf{T})}.$$

1. Lemma on the diameter of the ball. Let X be a normed space with unit ball BX and let $\hat{L}_{n+1}$ be an $(n+1)$ -dimensional subspace. Then

$$d_n(BX \cap \hat{L}_{n+1}, X) = 1.$$

In fact, let L_n be any n -dimensional subspace, and $L := \text{lin} \{\hat{L}_{n+1} \cup L_n\}$. The unit finite-dimensional ball $BX \cap L$ can be regarded as strictly convex (for it is easy to see that any finite-dimensional ball can be approximated arbitrarily closely by a strictly convex one). But then the best projection of $x \in S = \{\xi \in BX \cap \hat{L}_{n+1} \mid \|\xi\| = 1\}$ onto L_n is a continuous odd function

on the set S , a homeomorphic image of $\mathbf{S}^n$, under which diametrically opposite points are mapped to diametrically opposite points. Hence, by Borsuk's theorem, there is an element $\hat{x} \in S$ such that $d(\hat{x}, L_n, X) = \|\hat{x}\| = 1$. This proves the lemma.

Note that the complex version of this theorem is true as well.

2. **The upper bound** in both theorems follows immediately from 0.1 and 0.2.

3. **The lower bounds** follow from the lemma on the diameter of the ball: in Theorem 1 we must take $\hat{L}_{2n}$ as the space $S_{2n,r}$, and in Theorem 5 a) we take $\hat{L}_{n+1}$ as the space $\mathcal{P}_n$, and use Rolle's theorem in the first case, and the Bernstein-Zygmund inequality in the second.

8.6. **Theorem 2** of §3 (Ligon).

0.1. Kolmogorov's inequality (Theorem 3 of §1). 0.2. Rolle's theorem.

0.3. The interpolatory property of $S_{2n,r}$ on the system of points $\sigma_{n,r}$, where $\sigma_{n,r} = \{\pi k/n\}$ for r even, and $\{\pi k/n\} + \pi/2n$ for r odd ($|k| \leq n$) [17].

1. **Lemma.** Let $r \geq 2$ be a natural number, $x(\cdot) \in W_1(\mathbb{T})$, and $x(\tau_k) = 0$, $\tau_k \in \sigma_{n,r-1}$. Then $\|x(\cdot)\|_{L_1(\mathbb{T})} \leq n^{-r} K_r$.

Let $y_0(t) := \text{sign } x(t)$, $y_1(t) := \int_0^t y_0(\tau) d\tau + c$, where the constant c is

chosen so that the mean of $y_1(\cdot)$ over a period is zero, and let $y_r(\cdot)$ be the $(r-1)$ th periodic integral of $y_1(\cdot)$. Then, integrating by parts, we have (noting that $x(\pm\pi) = 0$)

$$(1) \quad \|x(\cdot)\|_{L_1(\mathbb{T})} \stackrel{\text{def}}{=} \int_{\mathbb{T}} |x(t)| dt = \int_{\mathbb{T}} x(t) y_0(t) dt = (-1)^r \int_{\mathbb{T}} x^{(r)}(t) y(t) dt.$$

By integration by parts it is easy to verify that $x^{(r)}(\cdot) \in S_{2n,r-1}^\perp(i)$. We interpolate $y_r(\cdot)$ by the spline $\xi_r(\cdot)$ from $S_{2n,r-1}$ at the points of the system $\sigma_{n,r-1}$ (see 0.3). Then (since $\|x^{(r)}(\cdot)\|_{L_1(\mathbb{T})} \leq 1$)

$$\|x(\cdot)\|_{L_1(\mathbb{T})} \stackrel{(1), (i)}{=} |(-1)^r \int_{\mathbb{T}} x^{(r)}(t) (y_r(t) - \xi_r(t)) dt| \leq \|y_r(\cdot) - \xi_r(\cdot)\|_{C(\mathbb{T})}.$$

We assume that $\|y_r(\cdot) - \xi_r(\cdot)\|_{C(\mathbb{T})} > n^{-r} K_r = \|\tilde{x}_{nr}(\cdot)\|_{C(\mathbb{T})}$ ($\tilde{x}_{nr}(\cdot)$ is the Euler spline, whose zeros coincide with $\sigma_{n,r-1}$). Then there is a point τ where $|y_r(\tau) - \xi_r(\tau)| > |\tilde{x}_{nr}(\tau)|$ ($\tilde{x}_{nr}^{(r)}(t) = \text{sign } \sin nt$). Let

$$\alpha := (y_r(\tau) - \xi_r(\tau)) / \tilde{x}_{nr}(\tau), \quad z(t) := y_r(t) - \xi_r(t) - \alpha \tilde{x}_{nr}(t).$$

We apply Rolle's theorem $r-1$ times, taking into account that $z(\cdot)$ has at least $2n+1$ zeros ($2n$ zeros in $\sigma_{n,r-1}$ and a zero at τ). However, the

function $z^{(r-1)}(t) = y_r^{(r-1)}(t) - \xi_r^{(r-1)}(t) - \alpha \tilde{x}_{nr}^{(r-1)}(t)$ is strictly monotonic on each

interval $[k\pi/n, (k+1)\pi/n]$, for there $|y_r^{(r)}(t)| \equiv 1$, $|\alpha \tilde{x}_{nr}^{(r)}(t)| = |\alpha| > 1$, and $\xi_r^{r-1}(\cdot)$ is constant. Hence the number of zeros of $z(\cdot)$ is at most $2n$. This contradiction proves the lemma.

2. Let $x(\cdot) \in W_1^{m+1}(\mathbf{T})$, $x(\tau_k) = 0$, $\tau_k \in \sigma_{n, k-1}$. Let

$$\Phi(t) := \int_{\mathbf{T}} x(t+\tau) \operatorname{sign} x^{(k)}(\tau) d\tau, \quad k := m+1-r \Rightarrow$$

$$(1) \quad \|\Phi(\cdot)\|_{C(\mathbf{T})} \stackrel{\text{def}}{\leq} \int_{\mathbf{T}} |x(t)| dt \stackrel{1.}{\leq} n^{-(m+1)} K_{m+1}.$$

Differentiating under the integral sign, we obtain

$$(2) \quad \Phi^{(j)}(t) = \int_{\mathbf{T}} x^{(j)}(t+\tau) \operatorname{sign} x^{(k)}(\tau) d\tau, \quad 1 \leq j \leq m+1, \Rightarrow$$

$$(3) \quad \|\Phi^{(k)}(\cdot)\|_{C(\mathbf{T})} \stackrel{\text{def}}{\geq} |\Phi^{(k)}(0)| \stackrel{(2)}{=} \int_{\mathbf{T}} |x^{(k)}(\tau)| d\tau \stackrel{\text{def}}{=} \|x^{(k)}(\cdot)\|_{L_1(\mathbf{T})},$$

$$(4) \quad \|\Phi^{(m+1)}(\cdot)\|_{L_\infty(\mathbf{T})} \stackrel{(2)}{\leq} \|x^{(m+1)}(\cdot)\|_{L_1(\mathbf{T})} \leq 1, \quad \Rightarrow (k := m+1-r)$$

$$\|x^{(k)}(\cdot)\|_{L_1(\mathbf{T})} \stackrel{(3)}{\leq} \|\Phi^{(k)}(\cdot)\|_{C(\mathbf{T})} \stackrel{0.1}{\leq} \dots$$

$$\stackrel{0.1}{\leq} K_{m+1-k} K_{\frac{m+1}{m+1}}^{\frac{k-m-1}{m+1}} \|\Phi(\cdot)\|_{C(\mathbf{T})}^{\frac{m+1-k}{m+1}} \|\Phi^{(m+1)}(\cdot)\|_{L_\infty(\mathbf{T})}^{\frac{k}{m+1}} \stackrel{(1), (4)}{\leq}$$

$$\stackrel{(1), (4)}{\leq} K_r K_{\frac{m+1}{m+1}}^{\frac{k-m-1}{m+1}} K_{\frac{m+1}{m+1}}^{\frac{m+1-k}{m+1}} n^{-\frac{r(m+1)}{m+1}} = n^{-r} K_r,$$

which by virtue of the duality relation

$$d(W_\infty^r(\mathbf{T}), S_{2n, m}, C(\mathbf{T})) \leq$$

$$\leq \sup \left\{ \|x^{(m+1-r)}(\cdot)\|_{L_1(\mathbf{T})} \mid x(\cdot) \in W_1^{m+1}(\mathbf{T}), x\left(\frac{\pi s}{n}\right) = 0, 0 \leq s \leq 2n-1 \right\}$$

proves the theorem. $\triangleright$

Note. The proof in §3 used Stein's method (see §8.9).

8.7. Proof of Theorem 4 of §3 (Ismagilov).

$\triangleleft$ 0.1. Let H be a Hilbert space, $\{f_i\}_{i=1}^s$ an orthonormal system, $x \in H$.

Then $d^2(x, \operatorname{lin} \{f_i\}_{i=1}^s, H) = \langle x, x \rangle - \sum_{i=1}^s \langle x, f_i \rangle^2$.

0.2. Parseval's equality.

0.3. Minkowski's inequality.

1. **Upper bound.** Let $x(\cdot) \in W_1^{(r)}(\mathbf{T}) \Rightarrow x(\cdot) = c + \varphi_r * h(\cdot)$, $\varphi_r(t) = \sum_{k=1}^{\infty} k^r \cos(kt - \pi r/2)$, $\|h(\cdot)\|_{L_1(\mathbf{T})} \leq 1$. Let

$$\varphi_{n-1,r}(t) := \sum_{k=1}^{n-1} k^r \cos(kt - \pi r/2), \quad y_{n-1}(\cdot) := c + \varphi_{n-1,r} * h(\cdot) \in \mathcal{T}_{n-1} \Rightarrow$$

$$(1) \quad \|x(\cdot) - y(\cdot)\|_{L_2(\mathbf{T})} \stackrel{0.3}{\leq} \|\varphi_r(\cdot) - \varphi_{n-1,r}(\cdot)\|_{L_2(\mathbf{T})} \stackrel{0.2}{=} \left(\sum_{k \geq n} k^{-2r} \right)^{1/2} =: \kappa_{nr}.$$

2. **The lower bound** is obtained (following Ismagilov) by the averaging method (see (§ 1.4).

Let L_{2n-2} be a subspace of dimension $2n-2$ in $L_2(\mathbf{T})$ and $\{f_i(\cdot)\}_{i=1}^{2n-2}$ an orthonormal basis in L_{2n-2} .

$$f_i(t) = \sum_{k \in \mathbf{Z}_+} (a_{ki} \cos kt + b_{ki} \sin kt) \Rightarrow$$

$$(2) \quad d^2(\varphi_r(\cdot - \tau), L_{2n-2}, L_2(\mathbf{T})) \stackrel{0.1}{=} \pi^{-1} \int_{\mathbf{T}} \varphi_r^2(t - \tau) d\tau - \\ - \sum_{i=1}^{2n-2} \left(\pi^{-1} \int_{\mathbf{T}} \varphi_r(t - \tau) f_i(t) dt \right)^2 \stackrel{0.2}{=} \sum_{k \in \mathbf{N}} k^{-2r} - \sum_{i=1}^{2n-2} \psi_i^2(\tau),$$

$$\psi_i(\tau) := \sum_{k \in \mathbf{N}} (a_{ki} + b_{ki}) k^r \cos k\tau + (a_{ki} - b_{ki}) k^r \sin k\tau.$$

Averaging over τ , we obtain

$$(3) \quad (2\pi)^{-1} \int_{\mathbf{T}} d^2(\varphi_r(\cdot - \tau), L_{2n-2}, L_2(\mathbf{T})) d\tau \stackrel{0.2, (2)}{=} \\ = \sum_{k \in \mathbf{N}} k^{-2r} - 2^{-1} \sum_{i=1}^{2n-2} \sum_{k \in \mathbf{N}} k^{-2r} (a_{ki}^2 + b_{ki}^2).$$

From the orthonormality of $\{f_i(\cdot)\}$ we find that $0 \leq \sum_{i=1}^{2n-2} (a_{ki}^2 + b_{ki}^2) \leq 2$ and

$$\sum_{i=1}^{2n-2} \sum_{k \in \mathbf{N}} (a_{ki}^2 + b_{ki}^2) = \sum_{i=1}^{2n-2} \|f_i(\cdot)\|_{L_2(\mathbf{T})}^2 = 2n-2. \quad \text{It is easy to see that the}$$

extremal problem $(\sum_{k \in \mathbf{N}} k^{-2r} \alpha_k \rightarrow \sup, 0 \leq \alpha_k \leq 2, \sum_{k \in \mathbf{N}} \alpha_k = 2n-2)$ has the

solution $\hat{\alpha}_1 = \dots = \hat{\alpha}_{n-1} = 2$; from this and (3) it follows that there is a $\tau \in \mathbf{T}$ for which

$$(4) \quad d(\varphi_r(\cdot - \hat{\tau}), L_{2n-2}, L_2(\mathbf{T})) \geq \kappa_{nr}.$$

But arbitrarily close to $\varphi(\cdot - \hat{\tau})$ there is a function $\xi(\cdot)$ in $W_1^r(\mathbf{T})$ such that $\int \xi(t) dt = 0$. Hence (since $W_1^r(\mathbf{T})$ contains constants) it follows at once that

$$\kappa_{nr} \leq d_{2n-1}(W_1^r(\mathbf{T}), L_2(\mathbf{T})) \stackrel{(1)}{\Rightarrow} d_{2n-1}(W_1^r(\mathbf{T}), L_2(\mathbf{T})) = \kappa_{nr}. \quad \triangleright$$

8.8. **Theorem 3 of §3** (Tikhomirov).

This will be proved by Zhensykbayev's method; the original proof (also quite simple) relied on the theory of extremal problems. We shall use:

0.1. Haar's uniqueness theorem and Chebyshev's theorem on alternance [8].

0.2. Borsuk's theorem.

1. We use the notations $\Theta^k := \partial B l_1^k := \{z \in \mathbf{R}^{k+1} \mid \sum_{i=1}^{k+1} |z_i| = 1\}$, $k = n - r$,

$$L_n := \text{lin} \{1, \dots, t^{n-1}\},$$

$$u(t, z) := \left\{ \text{sign } z_j, \quad t \in \Delta_j := \left[\sum_{s=0}^j |z_s|, \sum_{s=0}^{j+1} |z_s| \right], \quad 0 \leq j \leq k, \quad z_0 = 0, \right.$$

$$\left. x(t, z, \xi) = ((r-1)!)^{-1} \int_1^t (t-\tau)_+^{r-1} u(\tau, z) d\tau + \sum_{h=1}^r \xi_h t^{h-1} \right\}.$$

We intersect the set $x(\cdot, z, \xi)$ (the cylinder $\Theta^k \times \mathbf{R}^r$) with a sufficiently large ball $\alpha BC(I)$ (α sufficiently large so that this intersection Γ is homeomorphic to S^n). We construct the mapping $\Phi: \Gamma \rightarrow \mathbf{R}^n$ that sends $(z, \xi) \in \Gamma$ to the coefficients of the polynomial of best approximation to $x(\cdot, z, \xi)$ in L_n . By Haar's theorem, Φ is continuous and is obviously centrally-symmetric: $\Phi(-(z, \xi)) = -\Phi(z, \xi)$. By Borsuk's theorem there is a pair $(\hat{z}, \hat{\xi})$ such that the best approximating polynomial for $x(\cdot, \hat{z}, \hat{\xi})$ is zero, that is, $\|x(\cdot, \hat{z}, \hat{\xi})\|_{C(I)} = d(x(\cdot, \hat{z}, \hat{\xi}), L_n, C(I))$. But then, by the theorem on alternance, $x(\cdot, \hat{z}, \hat{\xi})$ has at least $(n+1)$ -alternance; however, it cannot have alternance greater than $n+1$ by Rolle's theorem (for otherwise $x^{(r)}(\cdot, \hat{z}, \hat{\xi})$ would have more than $n-r$ changes of sign, contradicting its definition). Thus its alternance is equal to $n+1$, so that (by Rolle's theorem again) $x^{(r)}(\cdot, \hat{z}, \hat{\xi})$ has $n-r$ sign changes. The uniqueness also is proved by Rolle's theorem.

2. The proof of the upper and lower bounds is carried out in a similar way to that used in Theorem 1 of §3 and the lemma of §8.6. Instead of $S_{2n, r-1}$ we need to take a similar space of splines $S_{2n, r-1}(x_{nr}(\cdot))$, for which the $(r-1)$ th derivative is piecewise constant on the same intervals as $x_{nr}^{(r)}(\cdot)$, to interpolate $x(\cdot)$ at the zeros of $x_{nr}(\cdot)$, and then proceed as in the lemma in §8.6. For the lower bound we take $S_{2n, r}(x_{nr}(\cdot))$ instead of $S_{2n, r}$ and again repeat the argument of Theorem 1 of §3.

8.9. **Proof of Theorem 1 of §4.**

0.1. Kolmogorov's inequality (Theorem 3 of §1).

0.2. Hölder's inequality.

0.3. Parseval's equality.

1. Case a) (Kolmogorov) was examined in §1. Let us prove Theorem 1 b) (Stein). For $x(\cdot) \in W_1^n(\mathbf{R})$ we put

$$\Phi(t) = \int_{\mathbf{R}} x(t+\tau) \operatorname{sign} x^{(h)}(\tau) d\tau \quad (\text{cf. } \S 8.6).$$

Then in the same way as we did in §8.6, we obtain

$$\|\Phi(\cdot)\|_{C^b(\mathbf{R})} \leq \|x(\cdot)\|_{L_1(\mathbf{R})} \quad (\text{i}), \quad \|\Phi^{(h)}(\cdot)\|_{C^b(\mathbf{R})} \geq |\Phi^{(h)}(0)| = \|x^{(h)}(\cdot)\|_{L_1(\mathbf{R})} \quad (\text{ii}),$$

$$\|\Phi^{(n)}(\cdot)\|_{L_\infty(\mathbf{R})} \leq \|x^{(n)}(\cdot)\|_{L_1(\mathbf{R})} \quad (\text{iii}),$$

$$\Rightarrow \|x^{(h)}(\cdot)\|_{L_1(\mathbf{R})} \stackrel{(\text{ii})}{\leq} \|\Phi^{(h)}(\cdot)\|_{C^b(\mathbf{R})} \stackrel{0.1}{\leq}$$

$$\stackrel{0.1}{\leq} K_{n-h} K_n^{\frac{h-n}{n}} \|\Phi(\cdot)\|_{C^b(\mathbf{R})}^{\frac{n-h}{n}} \|\Phi^{(n)}(\cdot)\|_{L_\infty(\mathbf{R})}^{\frac{h}{n}} \stackrel{(\text{i}), (\text{iii})}{\leq}$$

$$\stackrel{(\text{i}), (\text{iii})}{\leq} K_{n-h} K_n^{\frac{h-n}{n}} \|x(\cdot)\|_{L_1(\mathbf{R})}^{\frac{n-h}{n}} \|x^{(n)}(\cdot)\|_{L_1(\mathbf{R})}^{\frac{h}{n}}.$$

To prove that the inequality is precise we need to take the function $x_{n-1}(\cdot)$ in §1.3, and convert it to a function of compact support by smoothing it out far from the origin.

2. Case 1 b) of Hardy-Littlewood-Pólya is quite simple. Let $\hat{x}(\cdot)$ be the Fourier transform of $x(\cdot)$; then

$$\int_{\mathbf{R}} |x^{(h)}(\cdot)|^2 dt \stackrel{0.3}{=} a^2 \int_{\mathbf{R}} \sigma^{2h} |\hat{x}(\sigma)|^2 d\sigma \stackrel{\text{Id}}{=}$$

$$\stackrel{\text{Id}}{=} a^2 \int_{\mathbf{R}} |\hat{x}(\sigma)|^{2(1-h/n)} \sigma^{2h} |\hat{x}(\sigma)|^{2h/n} d\sigma \stackrel{0.2}{\leq}$$

$$\stackrel{0.2}{\leq} \left(a \int_{\mathbf{R}} |\hat{x}(\sigma)|^2 d\sigma \right)^{2(1-h/n)} \left(a \int_{\mathbf{R}} \sigma^{2n} |\hat{x}(\sigma)|^2 d\sigma \right)^{2h/n} \stackrel{0.3}{=}$$

$$\stackrel{0.3}{=} \|x(\cdot)\|_{L_1(\mathbf{R})}^{2(1-h/n)} \|x^{(n)}(\cdot)\|_{L_1(\mathbf{R})}^{2h/n}.$$

3. Case 1 d) (Taikov) is proved similarly. Here we need to estimate the quantity $|x^{(h)}(0)| = \left| a \int \sigma^h \hat{x}(\sigma) d\sigma \right|$ by the quantities $a \int |\hat{x}(\sigma)|^2 d\sigma$ and $a \int \sigma^{2n} |\hat{x}(\sigma)|^2 d\sigma$, which is done by a simple application of Hölder's inequality.

Theorem 2 is proved by standard methods of the theory of extremal problems. We shall omit it because of shortage of space.

8.10. *Proof of Theorems 1 and 2 of §5* (Stesin-Khodulev and Pich-Stesin).

◁ 0.1. **Lebesgue's theorem** ($F \in C(Bl_p^n, \mathbb{R}^n)$, $\|x - Fx\|_{l_p^n} < 1 \Rightarrow \exists \delta > 0$, $\forall y \in \mathbb{R}^n$, $\|y\|_{l_p^n} < \delta \exists x \in Bl_p^n : Fx = y$).

Corollary. $a_k(Bl_p^n, l_p^n) = 1$ for $k < n$.

0.2. Inequalities for means.

0.3. Monotonicity of diameters.

1. **Lemma** (Khodulev). Let Z be a linear space, $C \subset Z$; let X, Y be normed spaces over Z , $n_1, n_2 \in \mathbb{Z}_+$. Then

$$a_{n_1+n_2}(C, X) \leq a_{n_1}(C, Y) a_{n_2}(BY, X).$$

We restrict ourselves to an indication of the proof. Let $F_1 \in C(C, K_1)$, $\|x - F_1x\|_Y \leq a_{n_1}(C, Y) + \varepsilon$, $F_2 \in C(BY, K_2)$, $\|x - F_2x\|_X \leq a_{n_2}(BY, X) + \varepsilon$ (F_1, F_2 continuous, $\dim K_i = n_i$, $i = 1, 2$). Let

$$Fx := F_1x + \alpha_1 F_2(\alpha_1^{-1}(x - F_1x)),$$

$$\alpha_1 := a_{n_1}(C, Y), \quad K_3 := K_{n_1} + \alpha_1 K_{n_2}.$$

Then $F \in C(C, K_3)$, $\dim K_3 \leq n_1 + n_2$, $\|x - Fx\|_X \leq a_{n_1}(C, Y) a_{n_2}(BY, X) + A\varepsilon \Rightarrow \stackrel{0.1}{1} \leq a_{n_1+n_2}(Bl_p^n, l_p^n) \leq a_{n_1}(Bl_p^n, l_p^n) \times a_{n_2}(Bl_q^n, l_p^n)$. (i)

2. Lower bound.

a) $p \leq q$. Let $L_{k+1} := \text{lin}\{e_i\}_{i=1}^{k+1}$, and e_i the standard basis in $\mathbb{R}^n$. Let

$$x \in (k+1)^{1/q-1/p} Bl_q^n \cap L_{k+1} \Rightarrow \|x\|_{l_p^n} \stackrel{\text{def}}{=} \left(\sum_{i=1}^{k+1} |x_i|^p \right)^{1/p} \stackrel{0.2}{\leq} (k+1)^{1/p-1/q} \|x\|_{l_q^n} \stackrel{\text{def}}{\leq} 1 \Rightarrow$$

$$\Rightarrow (k+1)^{1/q-1/p} Bl_p^n \cap L_{k+1} \subset Bl_p^n \Rightarrow a_k(Bl_p^n, l_q^n) \stackrel{0.3}{\geq} a_k((k+1)^{1/q-1/p} Bl_p^n, l_q^n) \stackrel{0.4}{=}$$

$$\stackrel{0.1}{=} (k+1)^{1/q-1/p}. \quad (\text{ii})$$

$$\text{b)} \quad p \geq q. \quad a_k(Bl_p^n, l_q^n) \stackrel{(i)}{\geq} (a_{n-k-1}(Bl_q^n, l_p^n))^{-1} \stackrel{(ii)}{\geq} (n-k)^{1/q-1/p}.$$

3. Upper bound.

a) For $p \geq q$ we need to put $Fx = (x_1, \dots, x_k, 0, \dots, 0)$ and use 0.2. b) For $p \leq q$ we obtain the bound by Stesin's method. Let $x \in \mathbb{R}^n$, $(y_1, \dots, y_n) := \Pi x$, the rearrangement of $|x_i|$ in decreasing order. Let $\xi = \xi(x) := y_{n-k}$, $Fx = ((|x_1| - \xi(x))_+ \text{sign } x_1, \dots, (|x_n| - \xi(x))_+ \text{sign } x_n)$. It is easy to verify

that $F \in C(Bl_p^n, K)$, where K is a complex of dimension k . Let

$$\begin{aligned}
 N_1 &= \{i \mid |x_i| \geq \xi(x)\}, \\
 \delta &:= \sum_{i=1}^{n-k} y_i^p \Rightarrow 1 \geq \delta + \sum_{i=n-k+1}^n y_i^p \geq \delta + (k+1) \xi^p. \quad (\text{iii}) \\
 \|x - Fx\|_{l_q^n}^q &\stackrel{\text{def}}{=} \sum_{i \in N_1} |x_i|^q + \sum_{i \notin N_1} \xi^q = \sum_{i=1}^{n-k-1} y_i^q + (k+1) \xi^q \stackrel{\text{def}}{\leq} \\
 &\stackrel{\text{def}}{\leq} \xi^{q-p} \sum_{i=1}^{n-k-1} y_i^p + (k+1) \xi^q \stackrel{\text{def}}{=} \xi^{q-p} \delta + \xi^q (k+1) \stackrel{(\text{iii})}{\leq} \\
 &\stackrel{(\text{iii})}{\leq} \delta (1-\delta)^{(q-p)/p} (k+1)^{p-q} + (k+1) (1-\delta)^{1/p} (k+1)^{-q/p} \stackrel{\text{Id}}{=} \\
 &\stackrel{\text{Id}}{=} (k+1)^{(p-q)/p} (1-\delta)^{q/p} (\delta/(1-\delta) + 1) \stackrel{\text{Id}}{=} \\
 &\stackrel{\text{Id}}{=} (k+1)^{(p-q)/q} (1-\delta)^{(q-p)/p} \leq (k+1)^{(p-q)/q} \Rightarrow a_k(Bl_p^n, l_q^n) \leq (k+1)^{1/q-1/p}.
 \end{aligned}$$

This completes the proof of Theorem 1 a).

4. The mapping F in 3a) is a linear continuous finite-dimensional operator, and hence

$$\begin{aligned}
 (n-k)^{1/q-1/p} &\stackrel{2., 3}{=} a_k(Bl_p^n, l_q^n) \leq d_k(Bl_p^n, l_q^n) \leq \\
 &\leq \lambda_k(Bl_p^n, l_q^n) \leq \sup_{x \in Bl_p^n} \|x - Fx\|_{l_q^n} \stackrel{3.}{=} (n-k)^{1/q-1/p}.
 \end{aligned}$$

This proves the Pich-Stesin theorem.

8.11. *Proof of Kashin's lemma.* Here we prove a special case of Theorem 2 of §5 (see [34]).

Lemma. Let $1 \leq p \leq q \leq \infty$. Then $d_k(Bl_p^{2k}, l_q^{2k}) \asymp \{k^{1/q-1/p} \text{ in } V, 1 \text{ in } II, k^{1/q-1/2} \text{ in } III \cup IV\}$, where the regions II-V are shown in the figure of §5.

$\triangleleft 0.1$. $d_k(Bl_q^{2k}, l_q^{2k}) = 1$ (an obvious equality).

0.2. The Kolmogorov-Petrov-Smirnov-Mal'tsev-Stechkin theorem:
 $d_k(Bl_1^{2k}, l_2^{2k}) = 1/\sqrt{2}$.

0.3. Kashin's central lemma: $d_k(Bl_2^m, l_\infty^m) \leq Ck^{-1/2} \log^{1/2} \left(\frac{em}{k} \right)$. We shall not give a proof of this result here; for the original proof see [34]; recently a quite short proof due to Makovoz has appeared; like Kashin's original proof it is based on the averaging method.

0.4. An inequality for means: $1 \leq r \leq s \leq \infty \Rightarrow$

$$\|x\|_{l_s^n} \leq \|x\|_{l_r^n} \leq n^{1/r-1/s} \|x\|_{l_s^n} \Leftrightarrow n^{1/s-1/r} Bl_s^n \subset Bl_r^n \subset Bl_s^n.$$

0.5. Monotonicity and homogeneity of diameters.

1. Region II: $(1 \leq p \leq q, 1 \leq q \leq 2): 1 \ll d_h(Bl_1^{2h}, l_2^{2h})^{0.4, 0.5} \ll d_h(Bl_1^{2h}, l_q^{2h})^{0.4, 0.5} \ll d_h(Bl_q^{2h}, l_q^{2h})^{0.1} = 1$.

2. Region III $\cup$ IV: $(2 \leq q \leq \infty, 1 \leq p \leq 2): k^{1/q-1/2} \ll k^{1/q-1/2} d_h(Bl_1^{2h}, l_2^{2h})^{0.4, 0.5} = d_h(Bl_1^{2h}, l_q^{2h})^{0.4, 0.5} \ll k^{1/q} d_h(Bl_2^{2h}, l_q^{2h})^{0.4, 0.5} \ll k^{1/q} d_h(Bl_2^{2h}, l_\infty^{2h})^{0.3} \ll k^{1/q-1/2}$.

3. Region V $(2 \leq p \leq q)$.

$$\begin{aligned} k^{1/p-1/2} Bl_p^{2h} &\subset Bl_2^{2h} \text{ (i), } k^{1/q-1/p} Bl_q^{2h} \subset Bl_p^{2h} \subset Bl_q^{2h} \text{ (ii).} \\ k^{1/q-1/p} &= k^{1/q-1/p} d_h(Bl_q^{2h}, l_q^{2h})^{0.1} \\ &= d_h(k^{1/q-1/p} Bl_q^{2h}, l_q^{2h})^{(i), 0.5} \ll d_h(Bl_p^{2h}, l_p^{2h})^{(i), 0.5} \\ &\ll k^{-1/p+1/2} d_h(Bl_2^{2h}, l_q^{2h})^{2.} \ll k^{-1/p+1/2} \cdot k^{1/q-1/2} = k^{1/q-1/p}. \quad \triangleright \end{aligned}$$

8.12. *The Kolmogorov diameters of Sobolev classes.* Here we shall prove the following result (cf. Theorem 4 of §7).

Theorem. Let $1 < p, q < \infty, \alpha > 0$. Then $d_n(\dot{W}_p^\alpha(\mathbf{T}), L_q(\mathbf{T})) = d(n, \alpha, p, q) \asymp \{n^{-\alpha}, \alpha > 0 \text{ in I; } \alpha > 1/2 \text{ in V; } n^{-\alpha+1/p-1/q}, \alpha - 1/p + 1/q > 0 \text{ in II; } n^{-\alpha+1/p-1/2}, \alpha - 1/p > 0 \text{ in III} \cup \text{IV}\}$.

The proof uses the following facts. 0.1. The simplest properties of the diameter (monotonicity and homogeneity).

0.2. **Projection lemma** (Ismagilov). Let X be a normed space, $C \subset X$, L a subspace of X , P a continuous projection of X into L , $n \in \mathbf{Z}$. Then $d_n(C, X) \geq \|P\|^{-1} d_n(C, L)$.

0.3. **M. Riesz's projection theorem.** Let $x(\cdot) \in L_q(\mathbf{T})$, $1 < q < \infty$, $x(t) = \sum_{k \in \mathbf{Z}} x_k \exp(ikt)$, $P_{mn}x(t) := \sum_{k=m}^n x_k \exp(ikt)$. Then there exists $C > 0$ such that $\|P_{mn}x(\cdot)\|_{L_q(\mathbf{T})} \leq C \|x(\cdot)\|_{L_q(\mathbf{T})}$.

0.4. Marcinkiewicz's theorem on multipliers.

0.5. **The Marcinkiewicz-Zygmund theorem.** Let $1 < p < \infty$, $n \in \mathbf{Z}_+$, and μ_n the operator from $\mathcal{F}_n$ to $\mathbf{R}^{2n+1}$ mapping $x(\cdot) \in \mathcal{F}_n$ to the vector $((2n+1)^{-1} x(2\pi/(2n+1)), \dots, (2n+1)^{-1} x((2n+1)\pi/(2n+1)))$. Then there exist constants $c_p > 0$ and $C_p > 0$ such that

$$c_p \|\mu_n x(\cdot)\|_{l_p^{2n+1}} \leq \|x(\cdot)\|_{L_p(\mathbf{T})} \leq C_p \|\mu_n x(\cdot)\|_{l_p^{2n+1}} \text{ for all } x(\cdot) \in \mathcal{F}_n.$$

0.6. Kashin's lemma (8.11) and Kashin's central lemma (stated in 8.11).

0.7. Approximation to the class $\dot{W}_p^\alpha(\mathbf{T})$ by Fourier sums: if $\alpha > (1/p - 1/q)_+$, $1 < p, q < \infty$, then $\sup \{ \|x(\cdot) - S_n x(\cdot)\|_{L_q(\mathbf{T})} \mid x(\cdot) \in \dot{W}_p^\alpha(\mathbf{T}) \} \asymp n^{-\alpha + (1/p - 1/q)_+}$.

0.8. Embedding theorems for the classes $\dot{W}_p^\alpha(\mathbf{T})$. If $\alpha > 1/p - 1/q$, $1 < p \leq q < \infty$, then $\dot{W}_p^\alpha(\mathbf{T})$ is embedded in $\dot{W}_q^{\alpha - 1/p + 1/q}$, and if $p_1 < p_2$, then $\dot{W}_{p_1}^\alpha(\mathbf{T})$ is embedded in $\dot{W}_{p_2}^\alpha(\mathbf{T})$.

0.9. An inequality for diameters: let X be a normed space, $C \subset X$, $C = C_1 + C_2 + \dots + C_s + \dots$, $n_s \in \mathbf{Z}_+$ and $\sum_s n_s \leq n$. Then $d_n(C, X) \leq \sum_s d_{n_s}(C_s, X)$.

0.10. The Littlewood-Paley theorem.

Assertions 0.1, 0.2, and 0.9 are trivial. For the very important theorems of harmonic analysis, 0.3–0.5 and 0.10, see, for example, Zygmund's monograph on trigonometric series [36], and for 0.7 and 0.8 see [17].

1. Lower bound. Let L_{2n+1} denote the linear span of the exponentials $\exp(in\cdot)$, ..., $\exp(3in\cdot)$. Then

$$\begin{aligned} d_n(\dot{W}_p^\alpha(\mathbf{T}), L_q(\mathbf{T})) &\stackrel{0.1}{\gg} d_n(\dot{W}_p^\alpha(\mathbf{T}) \cap L_{2n+1}, L_q(\mathbf{T})) \stackrel{0.2, 0.3}{\gg} \\ &\gg d_n(\dot{W}_p^\alpha(\mathbf{T}) \cap L_{2n+1}, L_q(\mathbf{T}) \cap L_{2n+1}) \stackrel{0.4}{\gg} \\ &\gg d_n(n^{-\alpha} B L_p(\mathbf{T}) \cap L_{2n+1}, L_q(\mathbf{T}) \cap L_{2n+1}) \stackrel{0.5}{\gg} \\ &\gg n^{-\alpha + 1/p - 1/q} d_n(B l_p^{2n+1}, l_q^{2n+1}) \stackrel{0.6}{\asymp} d(n, \alpha, p, q). \end{aligned}$$

2. Lemma. Let $2 \leq q \leq \infty$, $\alpha > 1/2$. Then

$$d_n(\dot{W}_2^\alpha(\mathbf{T}), L_q(\mathbf{T})) \ll n^{-\alpha}.$$

$\triangleleft$ Without loss of generality we may assume that $n = 2^{\bar{s}}$. Let

$$\begin{aligned} x(\cdot) \in \dot{W}_2^\alpha(\mathbf{T}), \quad x(t) &= \sum_{k \in \mathbf{Z}_+} x_k \exp(ikt) = \sum_{s \in \mathbf{N}} (\delta_s x)(t), \quad \text{where } (\delta_s x)(t) := \\ &= \sum_{k \in \square_s} x_k \exp(ikt), \quad \square_s := \{k \mid 2^{s-1} \leq |k| \leq 2^s - 1\}. \quad \text{Let } W_2^\alpha(s) := \end{aligned}$$

$= \mathcal{T}_{\square_s} \cap \dot{W}_2^\alpha(\mathbf{T})$, $\mathcal{T}_{\square_s} = \text{lin}\{\exp(ik\cdot) \mid k \in \square_s\}$, $n^s = 2^s$ if $s = \bar{s}$, and $\{2^{s/\bar{s}}/2^{\beta(s-\bar{s})}\}$ if $s > \bar{s}$ (the number β , $0 < \beta < 1$, will be chosen a little later). Then, as it is easy to see, $\sum_s n_s \leq cn$, and thus

$$d_{cn}(\dot{W}_2^\alpha(\mathbf{T}), L_q(\mathbf{T})) \stackrel{0.9}{\ll} \sum_s d_{n_s}(W_2^\alpha(s), L_q(\mathbf{T})). \quad (\text{i})$$

Then we virtually repeat the calculation in para. 1:

$$\begin{aligned} d_{n_s}(W_2^\alpha(s), L_q(\mathbf{T})) &\ll d_{n_s}(W_2^\alpha(s), L_q(\mathbf{T}) \cap \mathcal{T}_{\square_s})^{0.4} \\ &\ll d_{n_s}(n_s^{-\alpha} B L_2(\mathbf{T}) \cap \mathcal{T}_{\square_s}, L_q(\mathbf{T}) \cap \mathcal{T}_{\square_s})^{0.5} \\ &\ll n_s^{-\alpha+1/2-1/q} d_{n_s}(B l_2^{2^{s+1}+1}, l_q^{2^{s+1}+1})^{0.5}, \quad (\text{ii}) \end{aligned}$$

whence, continuing (i) by taking (ii) into account, we obtain

$$\begin{aligned} \sum_s d_{n_s}(W_2^\alpha(s), L_q(\mathbf{T})) &\stackrel{(\text{ii})}{\ll} \sum_s n_s^{-\alpha+1/2-1/q} d_{n_s}(B l_2^{2^{s+1}+1}, l_q^{2^{s+1}+1})^{0.6} \ll (s = \bar{s} + k) \\ \sum_{k \in \mathbf{N}} 2^{\bar{s}(-\alpha+1/2-1/q)} \cdot 2^{k(-\alpha+1/2-1/q)} \cdot 2^{\bar{s}(1/q-1/2)} \cdot 2^{-\beta k(1/q-1/2)} \log \left(\frac{e 2^{\bar{s}+k} \cdot 2^{\beta k}}{2^{\bar{s}}} \right) &\ll \\ &\ll 2^{\bar{s}\alpha} \sum_{k \in \mathbf{N}} 2^{k(-\alpha+1/2-1/q-\beta(1/q-1/2))} (\beta(k+1) + \log_2 e) \ll n^\alpha. \end{aligned}$$

3. Upper bound. a) $\alpha > (1/p - 1/q)_+ \Rightarrow \text{I} \cup \text{II}$; b) $\alpha > 1/p - 1/2$, $(1/p, 1/q) \in \text{III} \cup \text{IV} \Rightarrow d_n(\dot{W}_p^\alpha(\mathbf{T}), L_q(\mathbf{T}))^{0.8, 0.1} \ll d_n(\dot{W}_p^{\alpha-1/p+1/2}(\mathbf{T}), L_q(\mathbf{T}))^{0.6} \ll n^{-\alpha+1/p-1/2}$; c) $\alpha > 0 \Rightarrow d_n(\dot{W}_p^\alpha(\mathbf{T}), L_q(\mathbf{T}))^{0.8} \ll d_n(\dot{W}_q^\alpha(\mathbf{T}), L_q(\mathbf{T}))^{0.6} \ll n^{-\alpha}$. $\triangleright$

Let us give a short summary of the methods involved in the proof of the results discussed.

The solution of the Kolmogorov-Nikol'skii problems most often uses various procedures of harmonic and asymptotic analysis (the Abel transformation, replacement of kernels by closer, but more simply constructed ones, and so on)—see §§1.1, 8.1, 8.4. In the area of Favard theory an important part is played by arguments like Favard's and also by modifications of Rolle's theorem (see §8.2). We note also the method of intermediate approximation (see §8.3) and the method of Σ -rearrangements developed by Korneichuk, which we have not touched on.

In exact calculations of diameters (and generally in exact solutions of extremal problems in approximation theory) we use topological methods (theorems of Borsuk, Lebesgue, and others)—see §§8.5–8.7, 8.10, the theory of splines (§§1.3 and 8.6), various averaging principles (§§1.4 and 8.8), different generalizations of Rolle's theorem (theory of *ET*-systems, representations of differential operators—see §8.2), the theory of equations of Sturm-Liouville type (§1.2), the theory of extremal problems (§§1.5, 8.9), a number of artificial methods (Kolmogorov's comparison theorem—see [11], Stein's method—see §§8.6–8.9—and others).

The investigation of the relation between the concepts of approximation and smoothness uses topological and geometric methods, fundamental theorems of harmonic analysis, the method of discretization, and many others—see §§1.6, 8.11, and 8.12.

Approximation theory is just one of twenty areas of mathematics (not counting applications) in which Kolmogorov worked.

This article has described the kind of role which his work has played in the development of approximation theory. I hope it has given the reader an idea of the enormous influence which Kolmogorov's creative work has had on the development of our science.

References

- [1] A.N. Kolmogorov, *Izbrannye trudy. Matematika i mekhanika* (Selected works. Mathematics and mechanics), Nauka, Moscow 1985. MR 87k:01082.
- [2] ———, *Teoriya veroyatnostei i matematicheskaya statistika. Izbrannye trudy* (Theory of probability and mathematical statistics. Selected works), Nauka, Moscow 1986. MR 88d:01043.
- [3] ———, *Teoriya informatsii i teoriya algoritmov* (Information theory and theory of algorithms), Nauka, Moscow 1987. MR 89a:01104.
- [4] J. Favard, Sur l'approximation des fonctions périodiques par des polynômes trigonométriques, C.R. Acad. Sci. Paris **203** (1936), 1122–1124.
- [5] N.I. Akhiezer and M.G. Krein, On best approximations by trigonometric sums to differentiable periodic functions, Dokl. Akad. Nauk SSSR **15** (1937), 107–111.
- [6] S.M. Nikol'skii, Approximation to periodic functions by trigonometric polynomials, Trudy Mat. Inst. Steklov. **15** (1945), 1–76. MR 7-435.
- [7] ———, *Kvadrurnye formuly* (Quadrature formulae), 2nd ed., Nauka, Moscow 1974. Zbl. **298.65017**.
- [8] N.I. Akhiezer, *Leksii po teorii approksimatsii* (Lectures on approximation theory), Nauka, Moscow 1965. MR 32 # 6108.
Translation: Vorlesungen über Approximationstheorie, Akademie-Verlag, Berlin 1967.
- [9] V.K. Dzyadyk, *Vvedenie v teoriyu ravnomernogo priblizheniya funktsii polinomami* (Introduction to the theory of uniform approximation of functions by polynomials), Nauka, Moscow 1977. MR 58 # 29579.
- [10] N.P. Korneichuk, *Ekstremal'nye zadachi teorii priblizheniya* (Extremal problems in approximation theory), Nauka, Moscow 1976. MR 56 # 6244.
- [11] ———, *Splainy v teorii priblizheniya* (Splines in approximation theory), Nauka, Moscow 1984. MR 87a:41001.
- [12] ———, *Tochnye konstanty v teorii priblizheniya* (Exact constants in approximation theory), Nauka, Moscow 1987. MR 89d:41038.
- [13] ———, A.A. Ligun, and V.G. Doronin, *Approximatsiya s ogranicheniyami* (Approximation with constraints), Naukova Dumka, Kiev 1982. MR 84j:41001.
- [14] A.I. Stepanets, *Ravnomernye priblizheniya trigonometricheskimi polinomami* (Uniform approximations by trigonometric polynomials), Naukova Dumka, Kiev 1981. MR 83k:42002.
- [15] ———, *Klassifikatsiya i priblizhenie periodicheskikh funktsii* (Classification and approximation of periodic functions), Naukova Dumka, Kiev 1987.
- [16] A.F. Timan, *Teoriya priblizheniya funktsii deistvitel'nogo peremennogo*, Fizmatgiz, Moscow 1960. MR 22 # 8257.
Translation: Theory of approximation of functions of a real variable, Pergamon Press, Oxford 1963. MR 33 # 465.
- [17] V.M. Tikhomirov, *Nekotorye voprosy teorii priblizhenii* (Some questions in approximation theory), Izdat. MGU, Moscow 1976. MR 58 # 6822.

- [18] N.P. Korneichuk, S.M. Nikol'skii and the development of research in the theory of approximation of functions in the USSR, *Uspekhi Mat. Nauk* **40**:5 (1985), 71-131. MR 87j:01040.
= Russian Math. Surveys **40**:5 (1985), 83-156.
- [19] S.B. Stechkin, Estimate of the remainder for Fourier series of differentiable functions, *Trudy Mat. Inst. Steklov.* **145** (1980), 126-151. MR 81e:42012.
= Proc. Steklov Inst. Math. **145** (1980), 139-166.
- [20] ———, Estimate of the remainder of a Taylor series for certain classes of analytic functions, *Izv. Akad. Nauk SSSR Ser. Mat.* **17** (1953), 461-472. MR 15-206.
- [21] K.I. Babenko, On best approximations to a class of analytic functions, *Izv. Akad. Nauk SSSR Ser. Mat.* **22** (1958), 631-640. MR 23 # A1048.
- [22] L.V. Taikov, On the best approximation in the mean of certain classes of analytic functions, *Mat. Zametki* **1** (1967), 155-162. MR 35 # 2047.
= Math. Notes **1** (1967), 104-109.
- [23] A.A. Ligun, Exact inequalities for upper bounds of seminorms on classes of periodic functions, *Mat. Zametki* **13** (1973), 647-654. MR 48 # 2633.
= Math. Notes **13** (1973), 389-393.
- [24] V.L. Velikin, On limiting constraints between approximations to periodic functions by splines and by trigonometric polynomials, *Dokl. Akad. Nauk SSSR* **258** (1984), 525-529. MR 83b:41014.
= Soviet Math. Dokl. **23** (1981), 540-544.
- [25] V.M. Alekseev, V.M. Tikhomirov, and S.V. Fomin, *Optimal'noe upravlenie* (Optimal control), Nauka, Moscow 1979. MR 81g:49001.
Translation: *Commande optimale*, Mir, Moscow 1982. MR 84m:49001.
- [26] A. Pinkus, n -widths of Sobolev spaces in L^p , *Constructive Approximation* **1** (1985), 15-62. MR 86d:41029.
- [27] ———, n -widths in approximation theory, *Ergebnisse der Math.* (3) **7**, Springer-Verlag, Berlin 1985. MR 86k:41001.
- [28] V.M. Tikhomirov, *Teoriya ekstremal'nykh zadach i teoriya priblizhenii* (Theory of extremal problems and approximation theory), in: *Approximation theory*, Banach Center Publication Vol. 4, 273-285, PWN, Warsaw 1979. MR 81g:41001.
- [29] A.P. Buslaev and V.M. Tikhomirov, Some problems of nonlinear analysis and approximation theory, *Dokl. Akad. Nauk SSSR* **283** (1985), 13-18. MR 86j:34023.
= Soviet Math. Dokl. **32** (1985), 4-8.
- [30] R.S. Ismagilov, On n -dimensional diameters of compact sets in Hilbert space, *Funktsional. Anal. i Prilozhen.* **2**:2 (1968), 32-39. MR 37 # 5682.
= Functional Anal. Appl. **2** (1968), 125-132.
- [31] S.B. Stechkin, Best approximations to linear operators, *Mat. Zametki* **1** (1967), 137-148. MR 35 # 2051.
= Math. Notes **1** (1967), 91-99.
- [32] R.S. Ismagilov, Diameters of sets in normed linear spaces and approximation to functions by trigonometric polynomials, *Uspekhi Mat. Nauk* **29**:3 (1974), 161-178. MR 53 # 11284.
= Russian Math. Surveys **29**:3 (1974), 169-186.
- [33] B.S. Kashin, Diameters of some finite-dimensional sets and classes of smooth functions, *Izv. Akad. Nauk SSSR Ser. Mat.* **41** (1977), 234-251. MR 58 # 1891.
= Math. USSR-Izv. **11** (1977), 317-333.
- [34] K.I. Babenko, *Osnovy chislennogo analiza* (Foundations of numerical analysis), Nauka, Moscow 1986. MR 88g:65001.

- [35] V.N. Temlyakov, Approximation to functions with bounded mixed derivative, *Trudy Mat. Inst. Steklov.* **178** (1986), 3-112.
= *Proc. Steklov Inst. Math.* **178** (1986).
- [36] A. Zygmund, *Trigonometric series*, 2nd ed. Cambridge University Press, Cambridge 1959. MR **21** # 6498.
Translation: *Trigonometricheskie ryady*, Mir, Moscow 1965.
- [37] G. Pólya and T. Szegő, *Problems and theorems in analysis*, 2 vols., Grundlehren Math. Wiss. **193**, **216**, Springer-Verlag, Berlin-New York 1972, 1976 (translated from the German). MR **49** # 8782, **53** # 2.
Russian translation: *Zadachi i teoremy iz analiza*, GITTL, Moscow 1956.
- [38] M.G. Krein, On the theory of best approximations to functions, *Dokl. Akad. Nauk SSSR* **18** (1938), 245-249.

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